CSE 311:
Data Communication

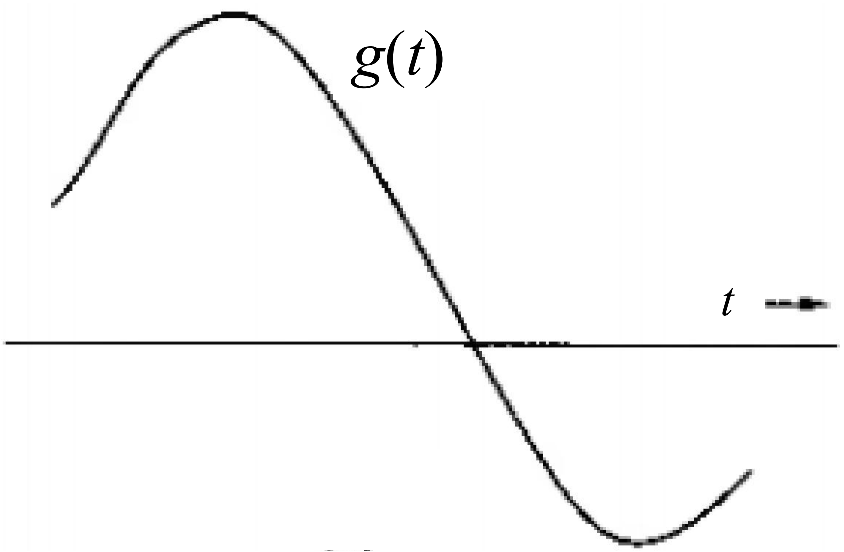
Instructor:
Dr. Md. Monirul Islam

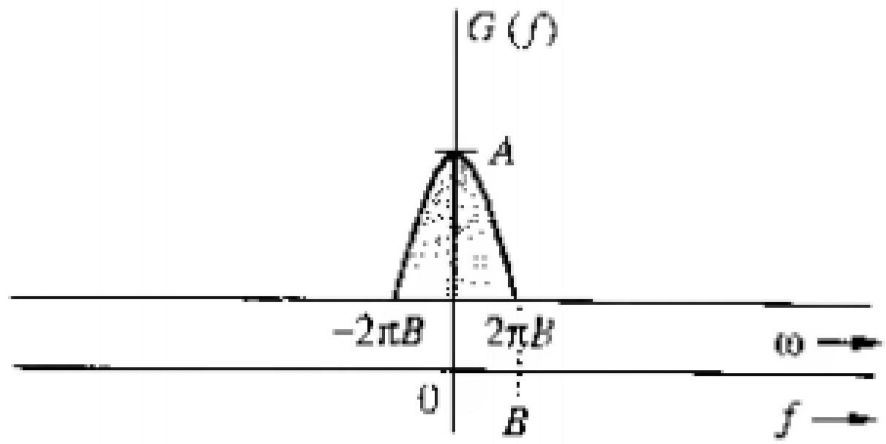
Sampling
and
Analog-to-Digital Conversion

Sampling Theorem

Review

Let a message signal $g(t)$ band limited to B Hz





$G(f) = 0$ for all $|f| > B$

The signal $g(t)$ can be reconstructed exactly from its discrete samples taken at a rate of f_s Hz, where $f_s \geq 2B$

Sampling Theorem

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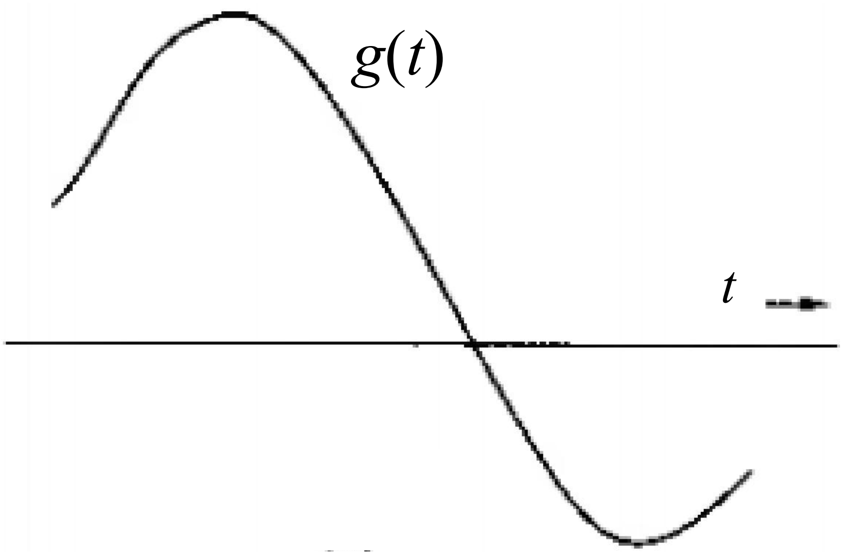
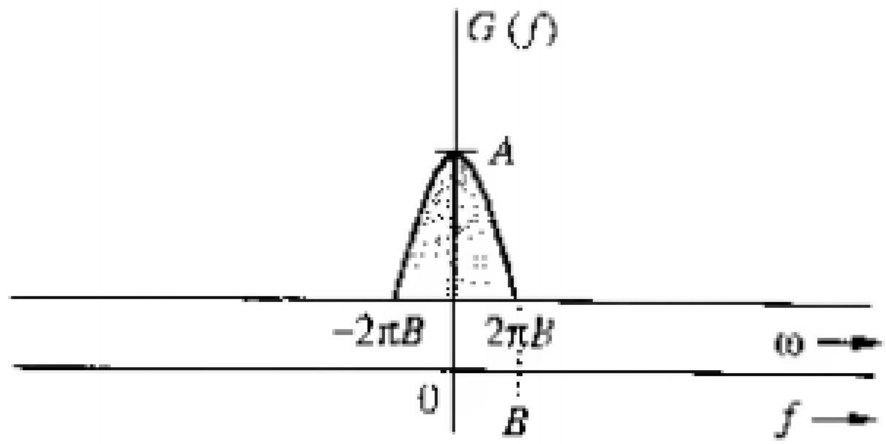
Let a message signal $g(t)$ band limited to B Hz





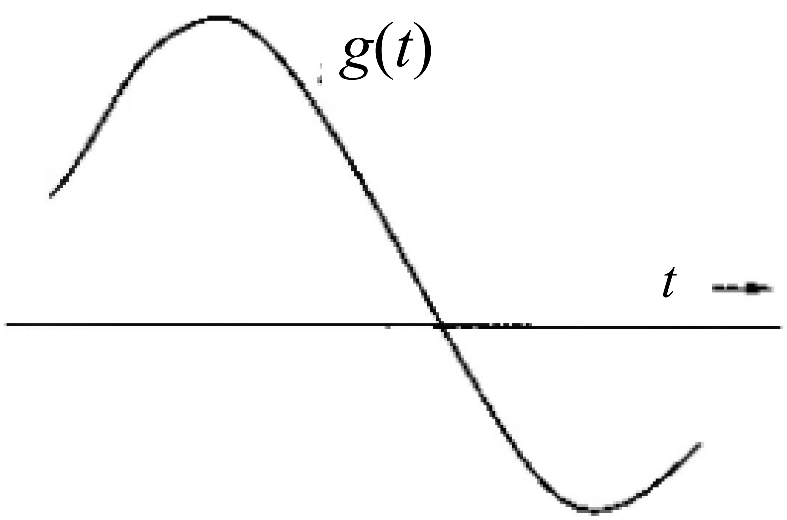
$G(f) = 0$ for all $|f| > B$

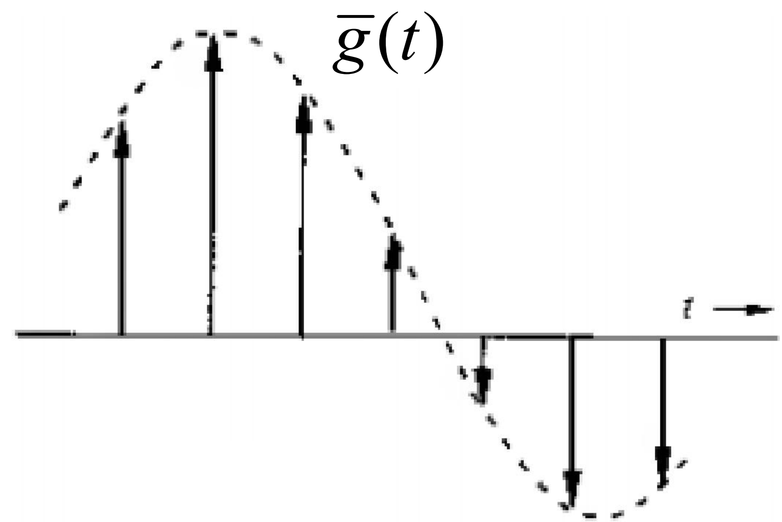
For recovery, the minimum sampling $f_s = 2B$ Hz

Sampling Theorem

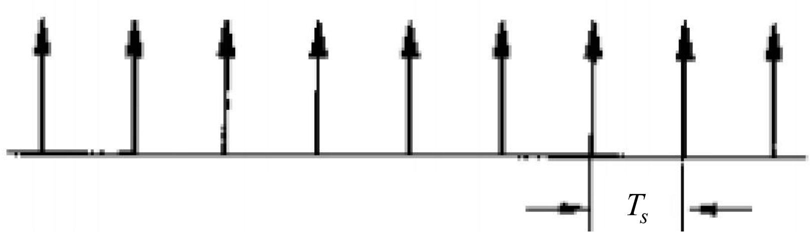
Review

Uniform sampling $g(t)$ at a rate f_s





$$\delta_{T_s}(t) = \sum_n \delta(t - nT_s)$$



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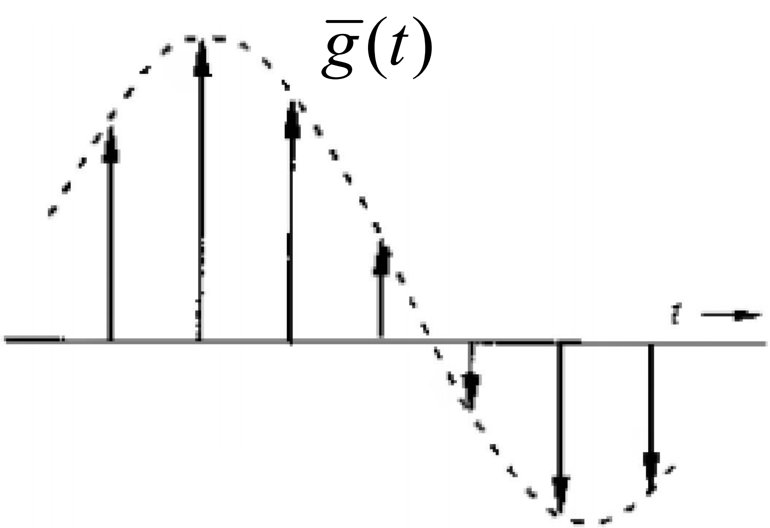


$$\phi(t)\delta(t) = \phi(0)\delta(t)$$

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$

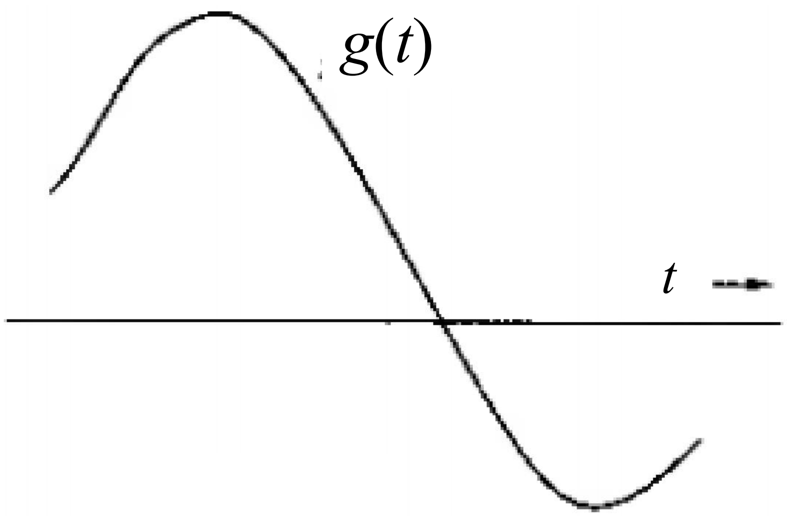
$\bar{g}(t)$ is multiplication of $g(t)$ and impulse train $\delta_{T_s}(t)$ and can be written as

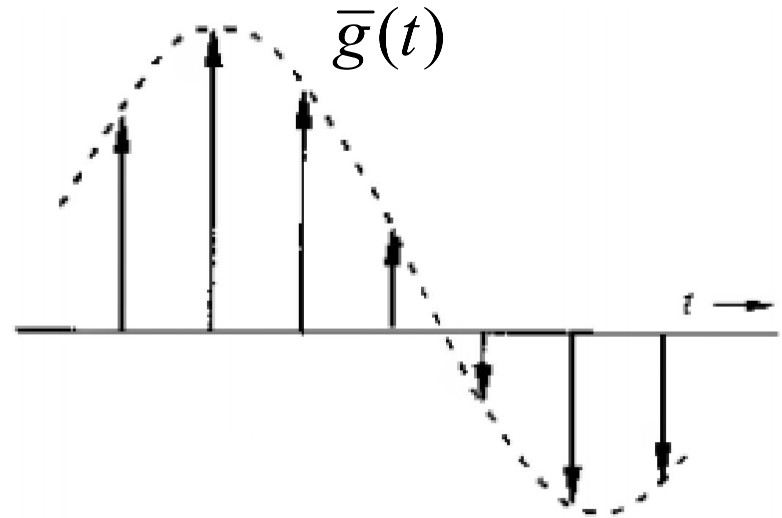
$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$

Sampling Theorem

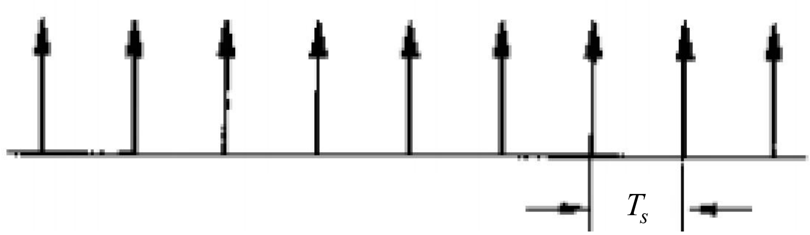
Review

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$





$$\delta_{T_s}(t) = \sum_n \delta(t - nT_s)$$



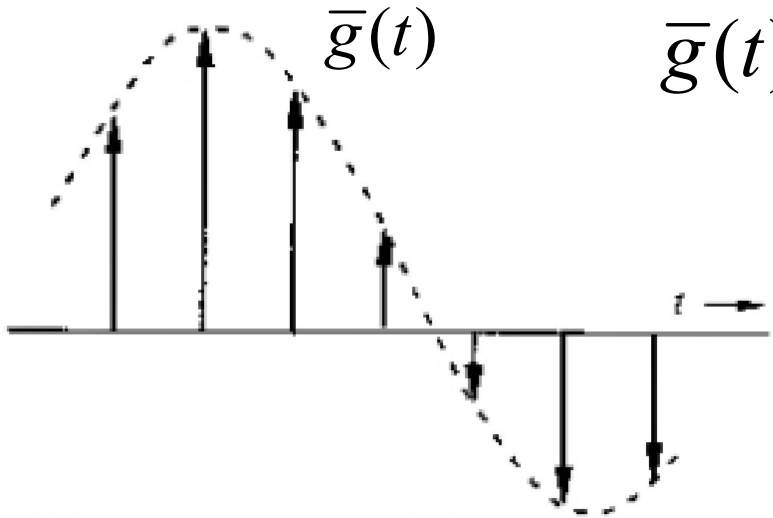
Alternate Representation

$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_n e^{jn2\pi f_s t}$$

Sampling Theorem

Review

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \frac{1}{T_s}\sum_n g(t)e^{jn2\pi f_s t}$$

$$= \frac{1}{T_s}\sum_n g(t)e^{j2\pi n f_s t}$$

$\bar{g}(t)$ is multiplication of $g(t)$ and impulse train $\delta_{T_s}(t)$ and can be written as

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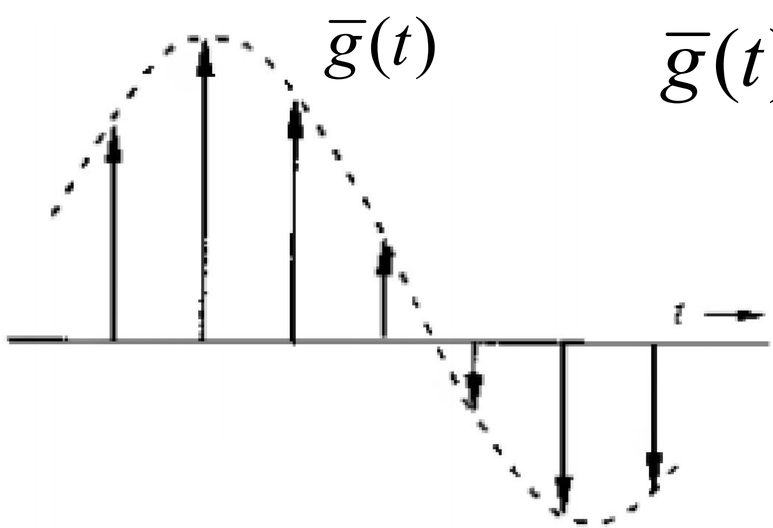


$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \frac{1}{T_s}\sum_n g(t)e^{jn2\pi f_s t}$$

$$= \frac{1}{T_s}\sum_n g(t)e^{j2\pi n f_s t}$$

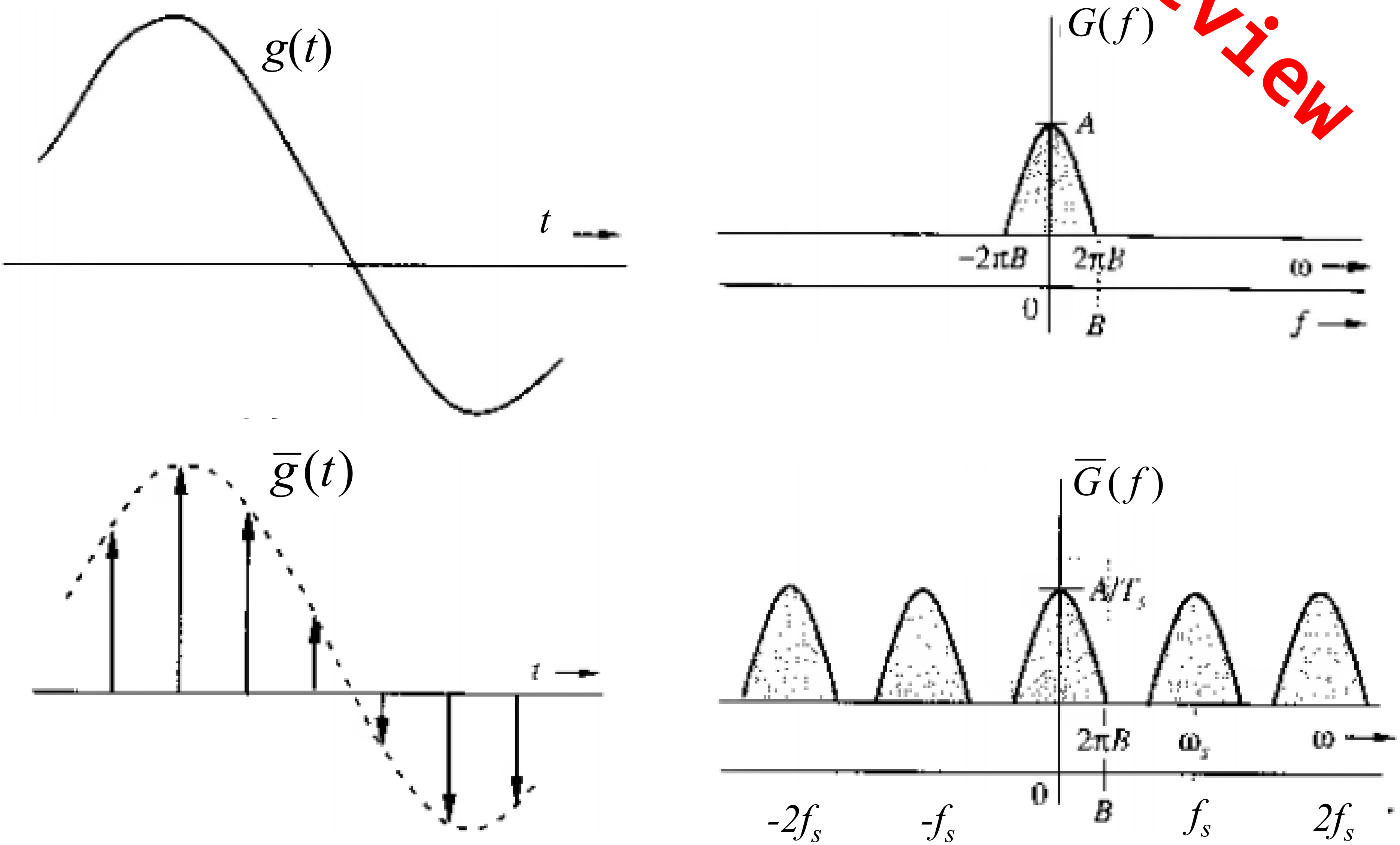
FT of $\bar{g}(t)$ is defined by

$$\bar{G}(f) = \Im[\bar{g}(t)] = \frac{1}{T_s}\sum_n G(f - nf_s)$$

$\bar{G}(f)$ consists of $G(f)$ scaled by f_s repeated periodically with period f_s

Sampling Theorem

Review



Sampling Theorem

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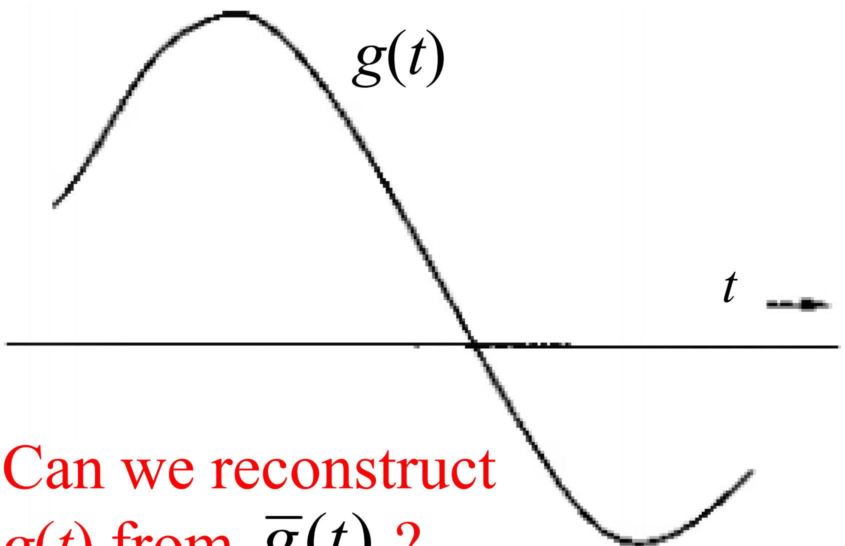
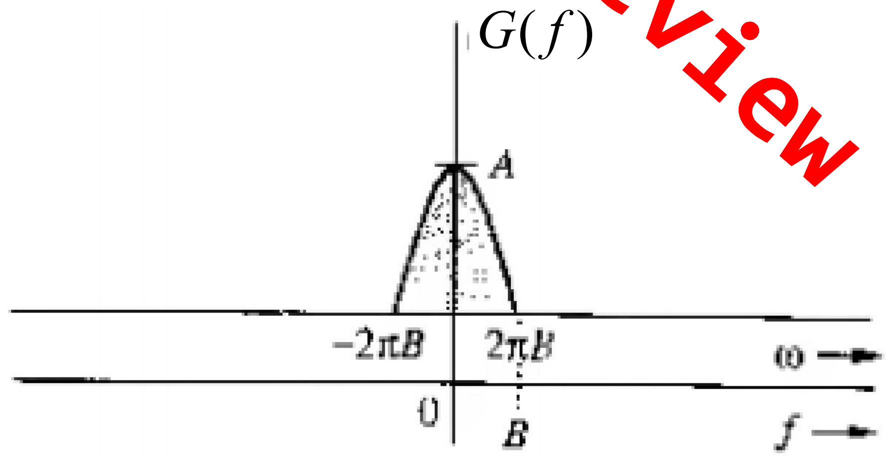
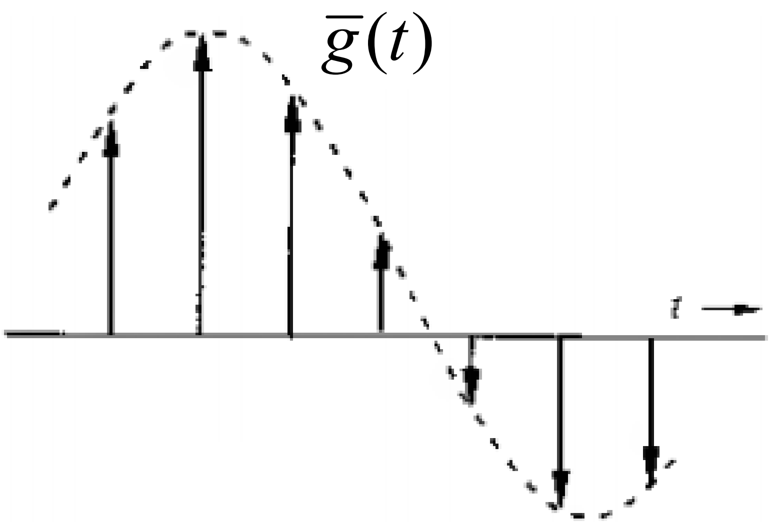
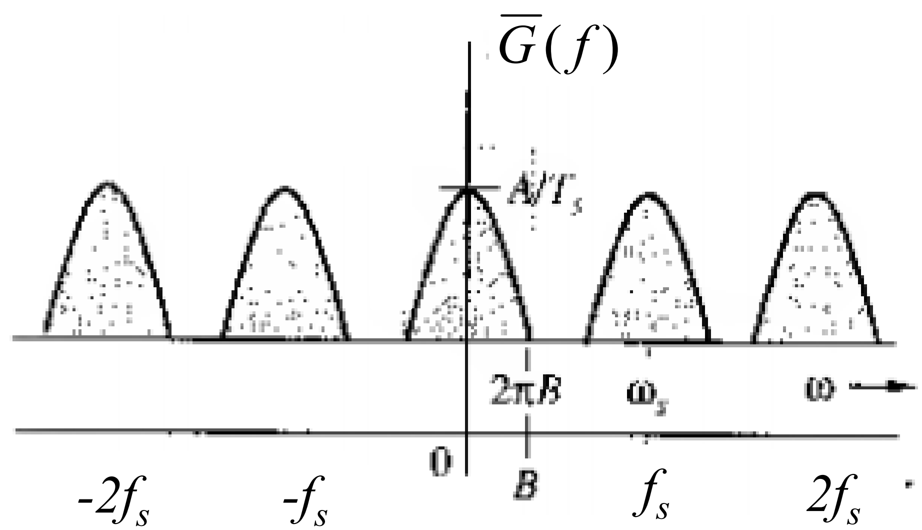




Can we reconstruct $g(t)$ from $\overline{g}(t)$?





Sampling Theorem

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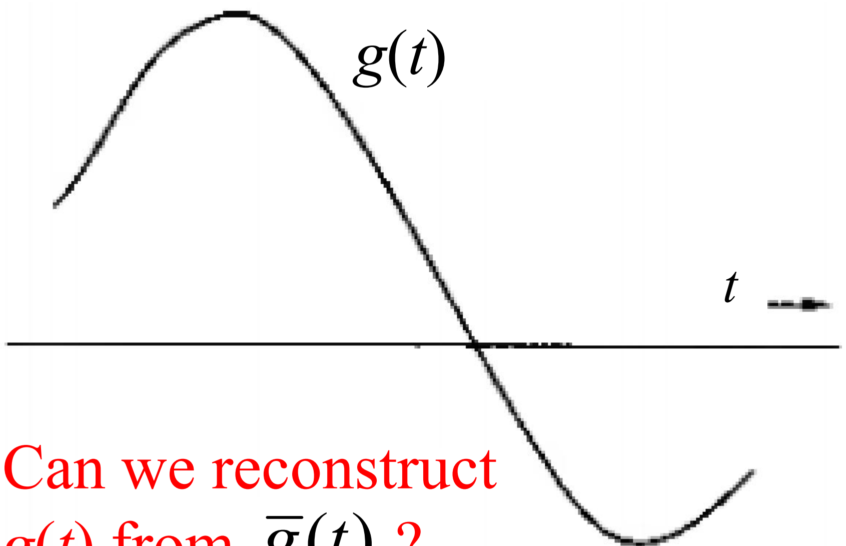

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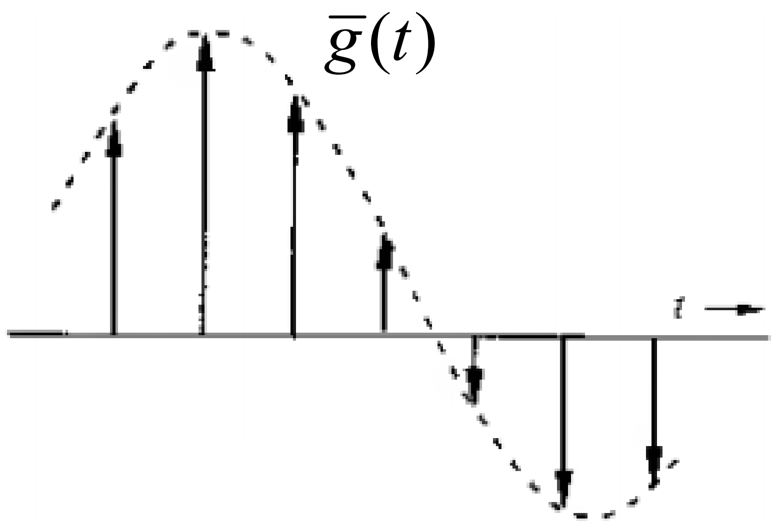
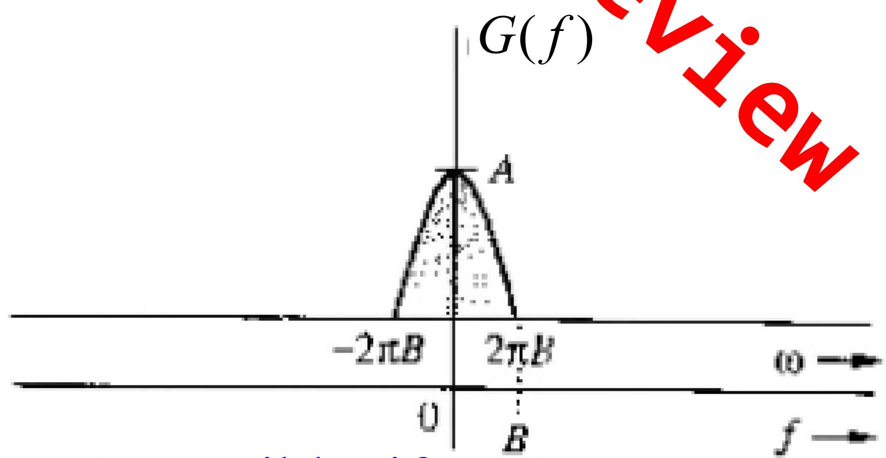
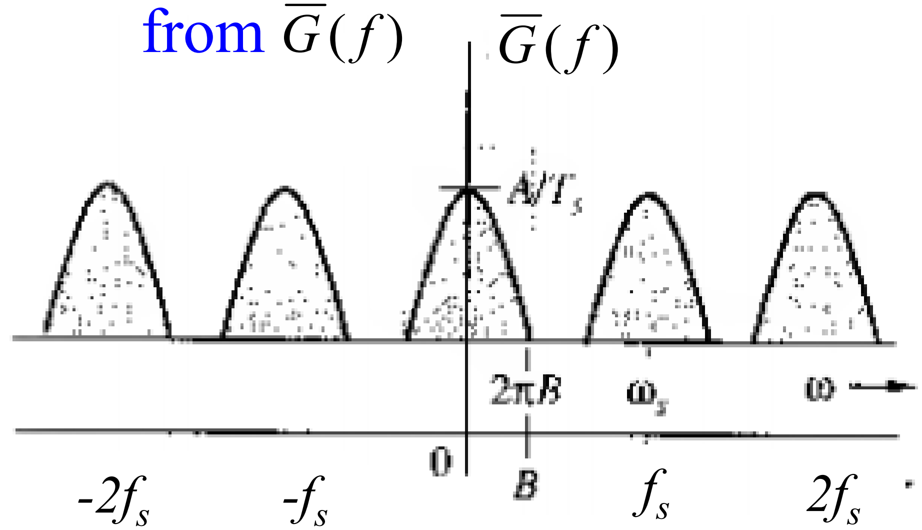




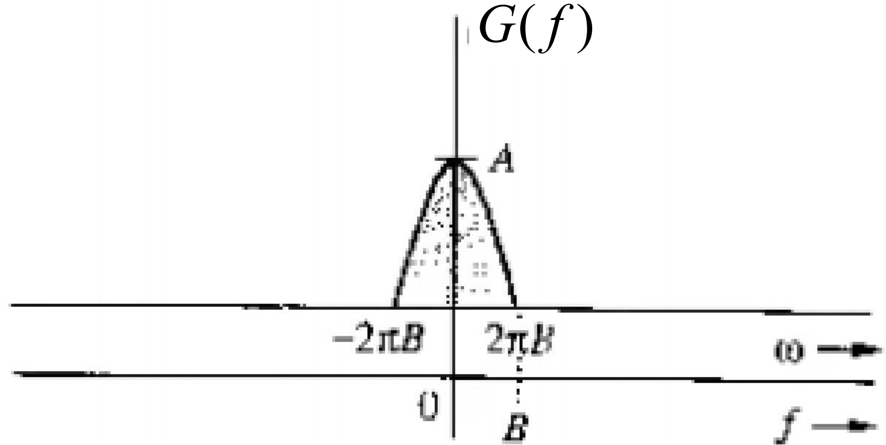
Possible, if we can we reconstruct $G(f)$ from $\overline{G}(f)$



Sampling Theorem

- We have to isolate one cycle from $\overline{G}(f)$
- This is possible when cycles are non-overlapping, this means

$f_s > 2B$



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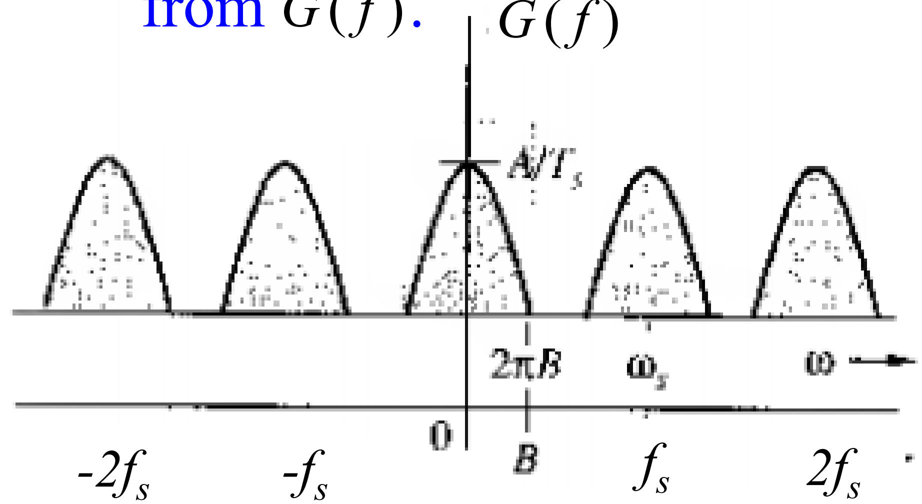

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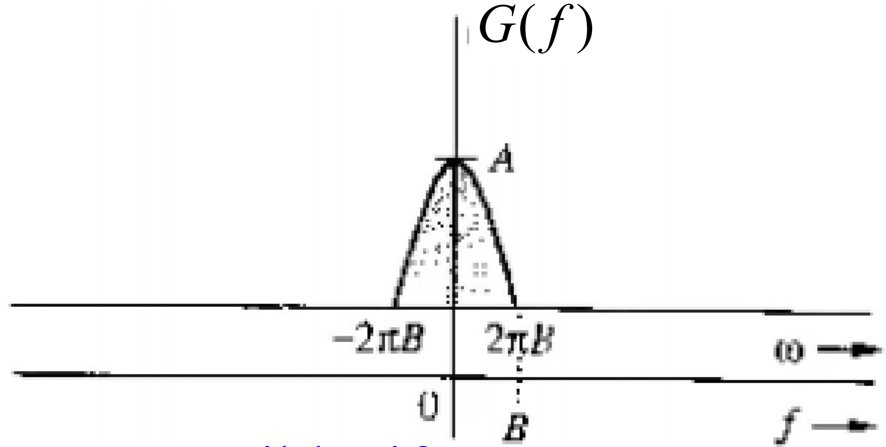
Alternately,

Sampling interval of $g(t)$, $T_s < \dfrac{1}{2B}$



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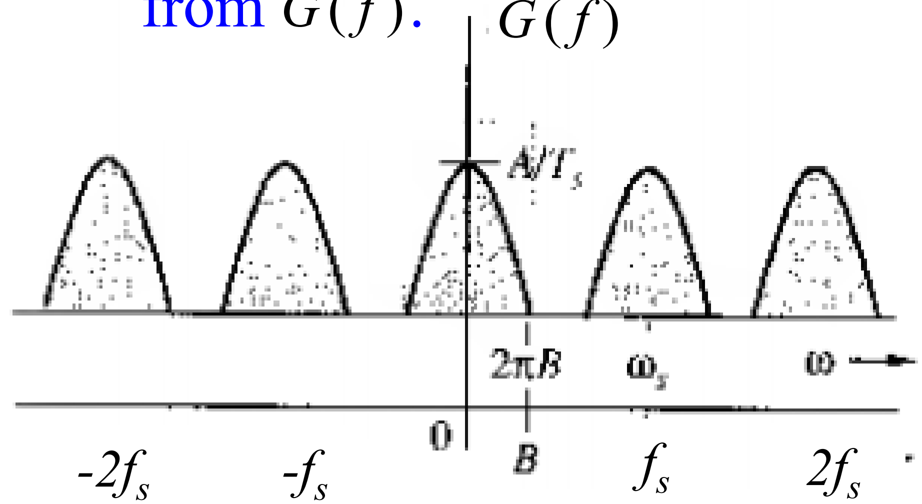
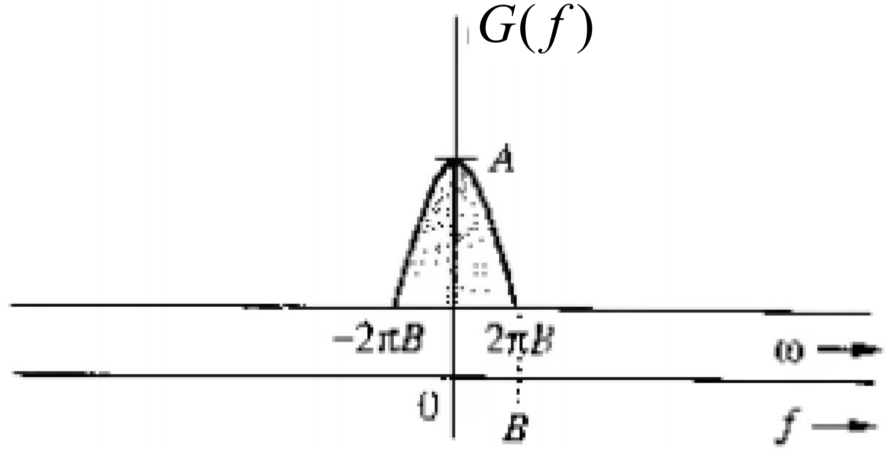


Sampling Theorem



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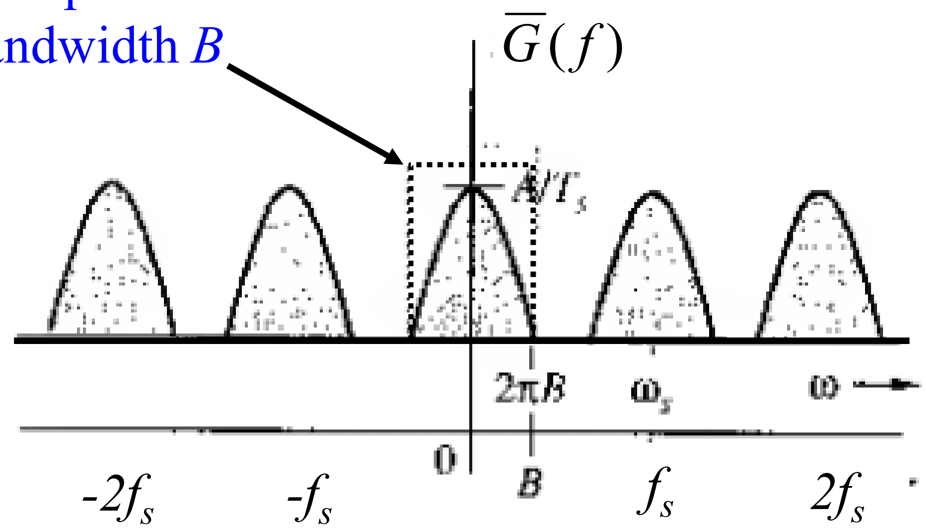
Alternately,

Sampling interval of *g(t)*, $T_s < \dfrac{1}{2B}$

Use a Ideal lowpass filter of bandwidth *B* to isolate one cycle.

Low pass filter of bandwidth *B*



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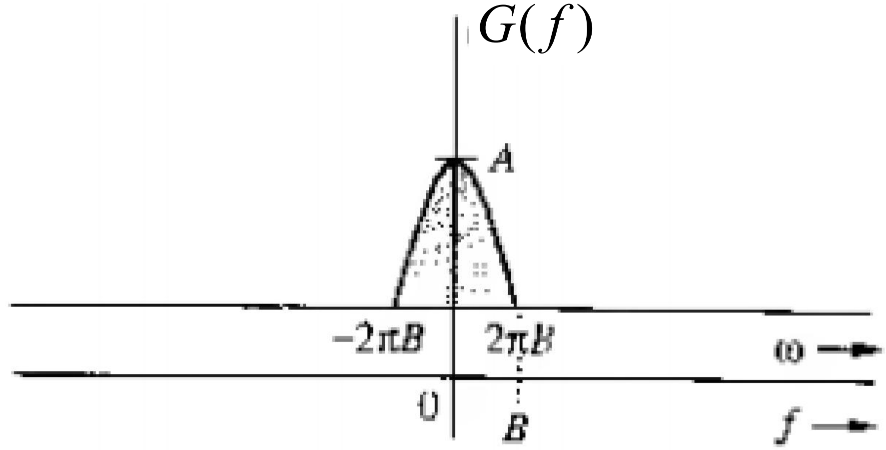
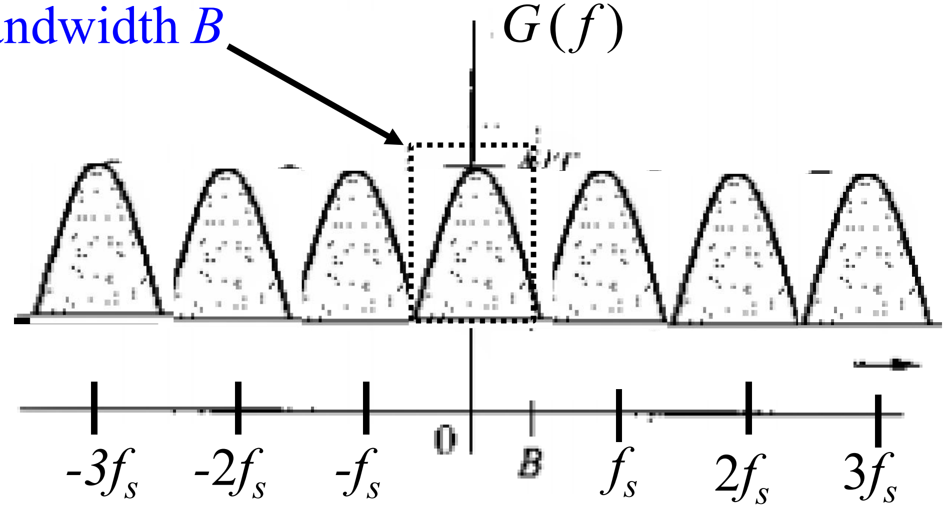


Minimum sampling rate can be

$f_s = 2B$

This is Nyquist sampling rate

Low pass filter of bandwidth B



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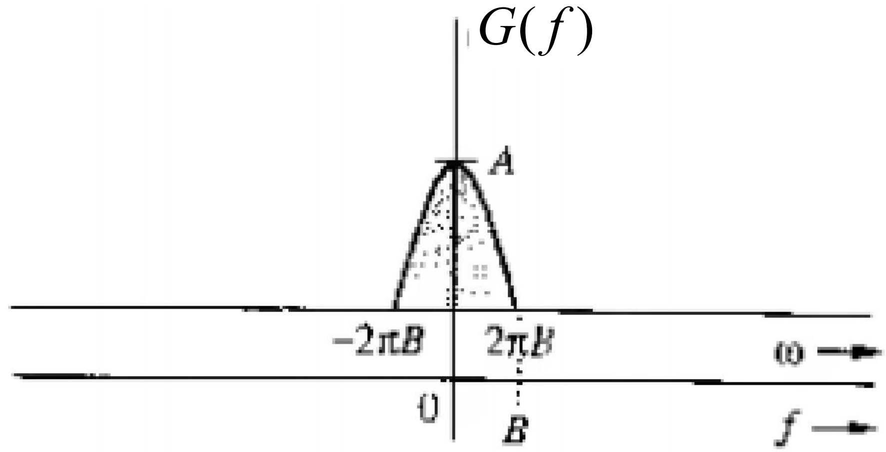

Maximum sampling interval is

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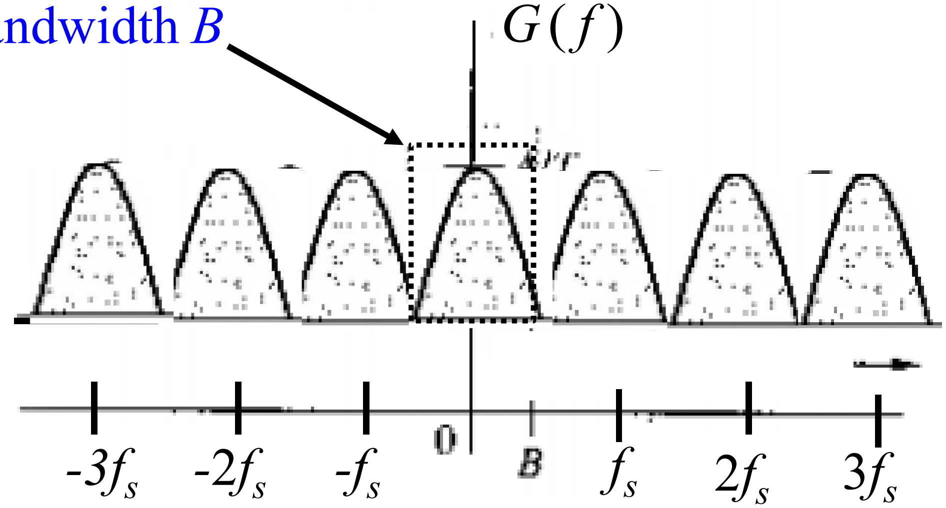
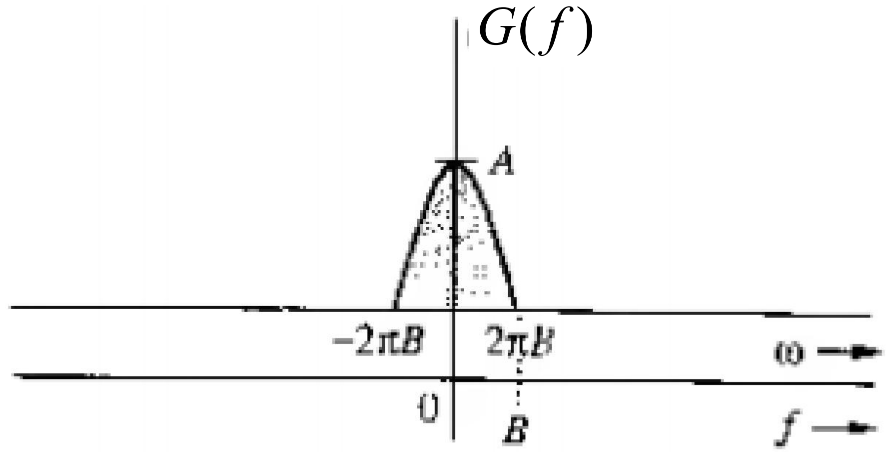
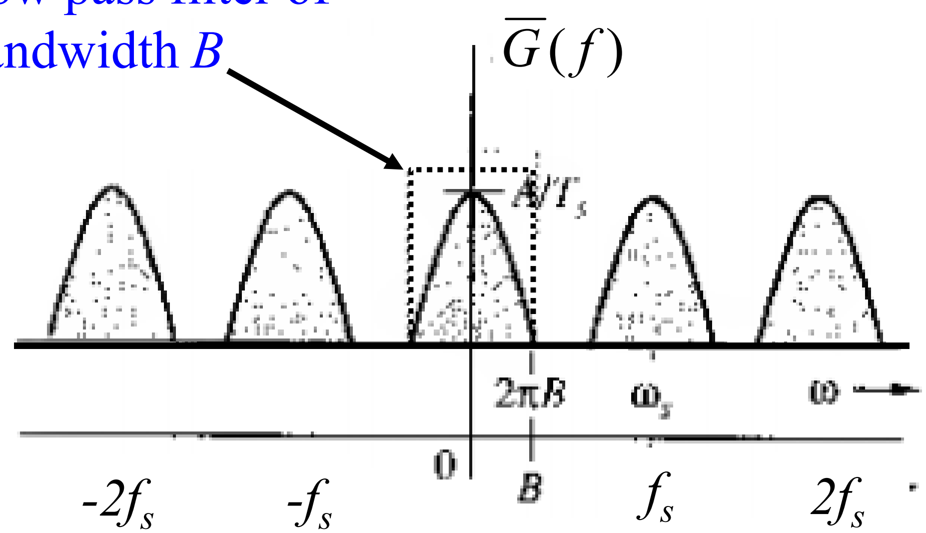


Sampling Theorem

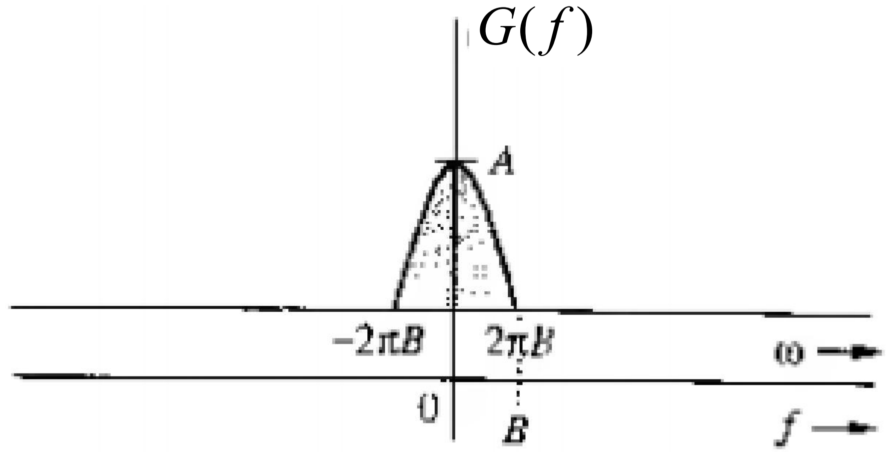
- We have to isolate one cycle from $\overline{G}(f)$
- For a general signal $g(t)$

$f_s > 2B$





Sampling Theorem

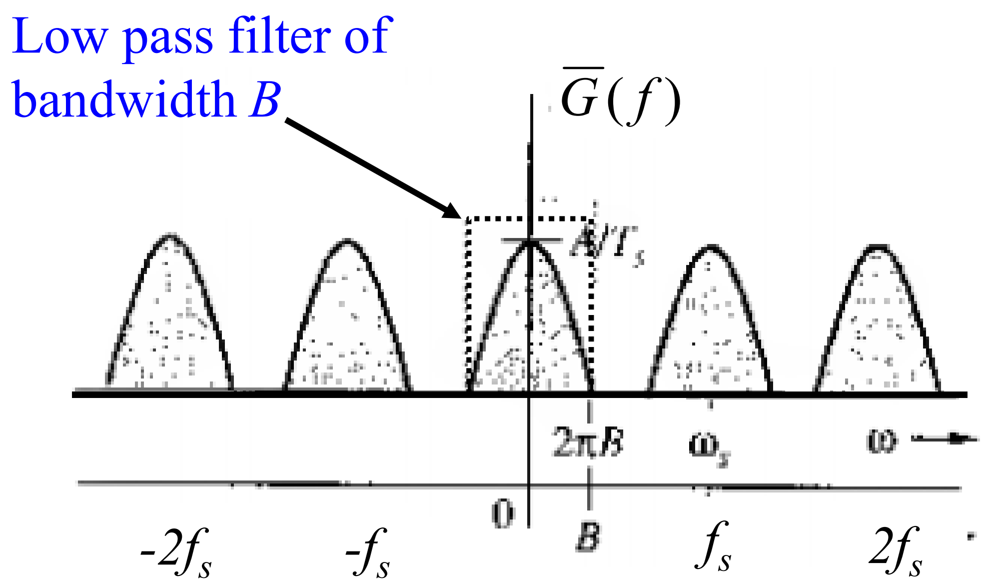


- We have to isolate one cycle from $\overline{G}(f)$
- For a general signal $g(t)$

  $f_s > 2B$

- If there is no impulse at ±B, we can use $f_s = 2B$. Therefore,

  $f_s \geq 2B$



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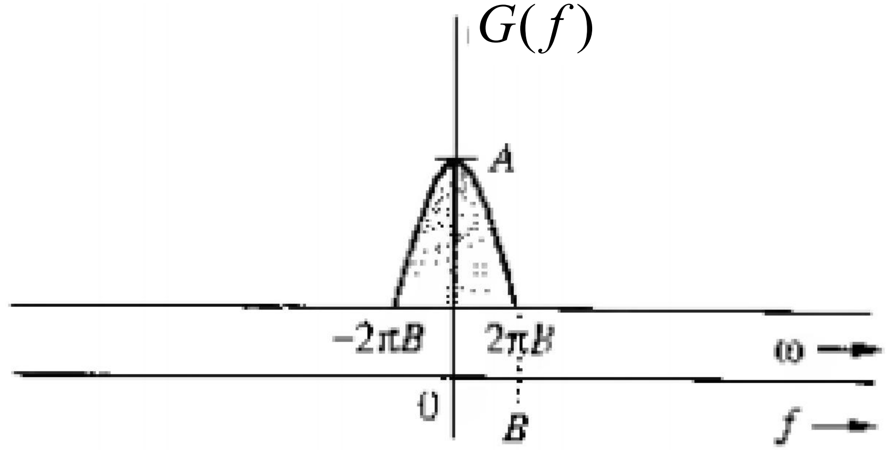

- We have to isolate one cycle from $\overline{G}(f)$
- For a general signal $g(t)$

$f_s > 2B$

- If there is no impulse at $\pm B$, we can use $f_s = 2B$. Therefore,

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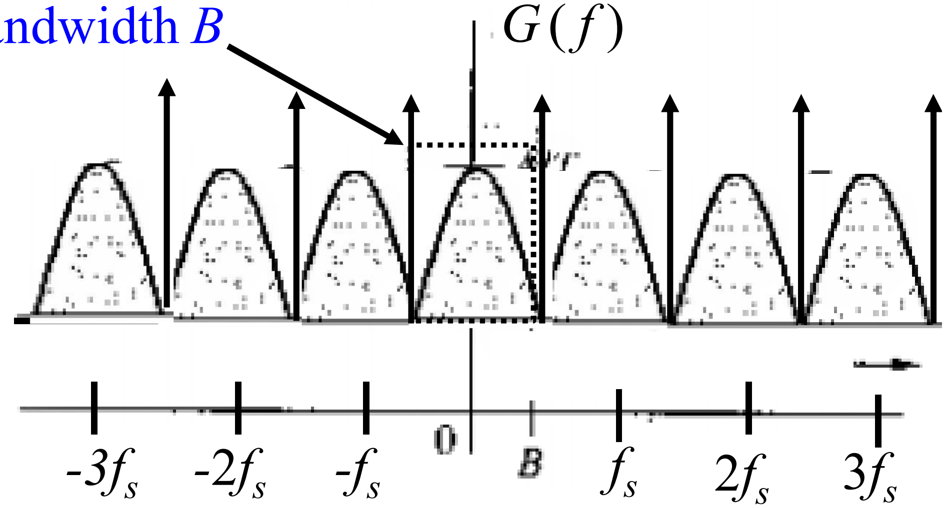
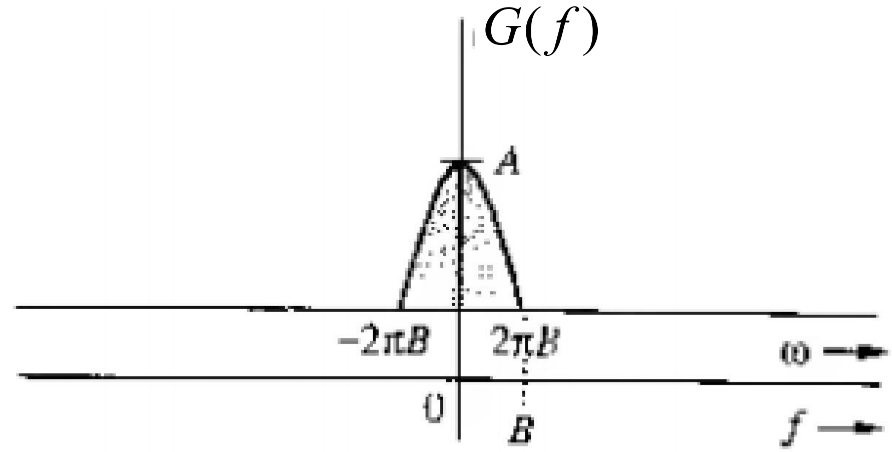
- If there is IMPULSE at $\pm B$, we can NOT use $f_s = 2B$, because overlap will occur. Therefore, $f_s > 2B$

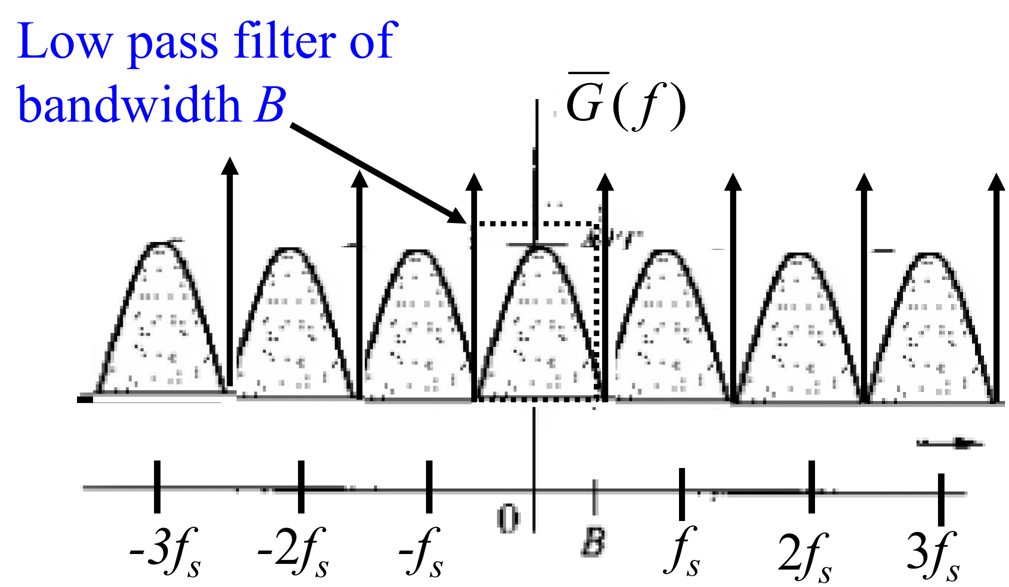
Low pass filter of bandwidth B



Sampling Theorem

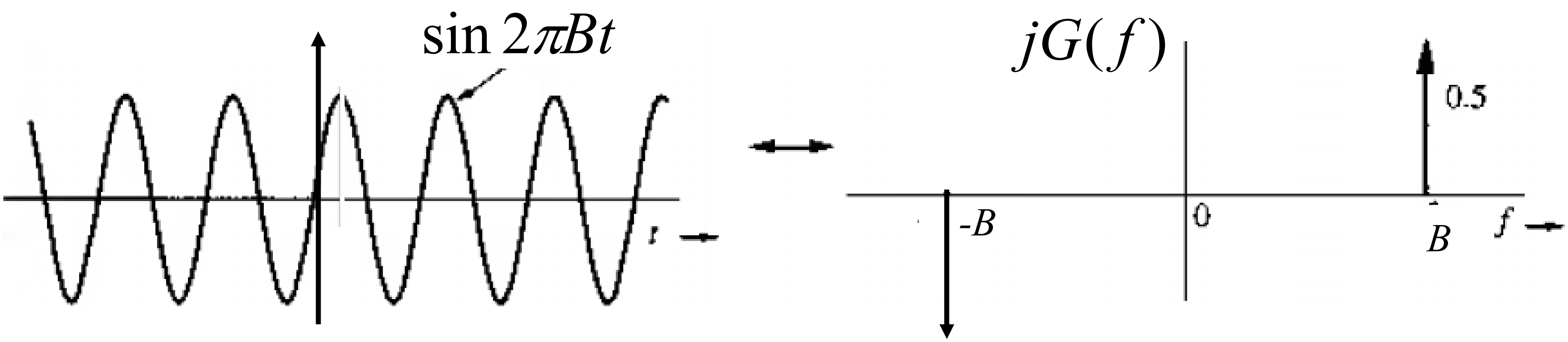
- In this case we cannot isolate one cycle from $\overline{G}(f)$ if $f_s = 2B$





Sampling Theorem

- In this case we cannot isolate one cycle from $\overline{G}(f)$ if $f_s = 2B$
- Example of sine wave of frequency B



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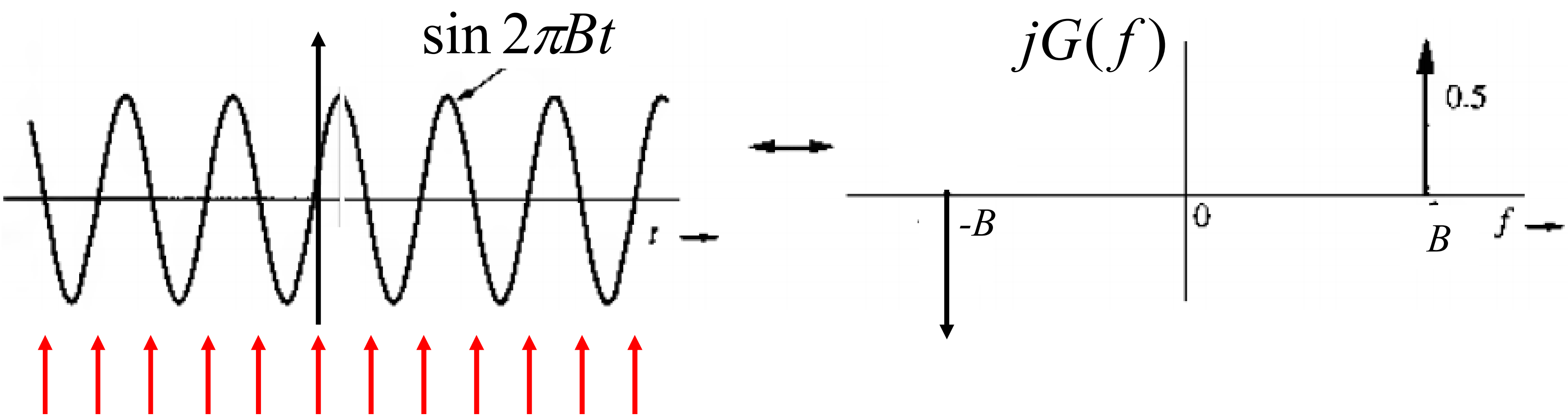


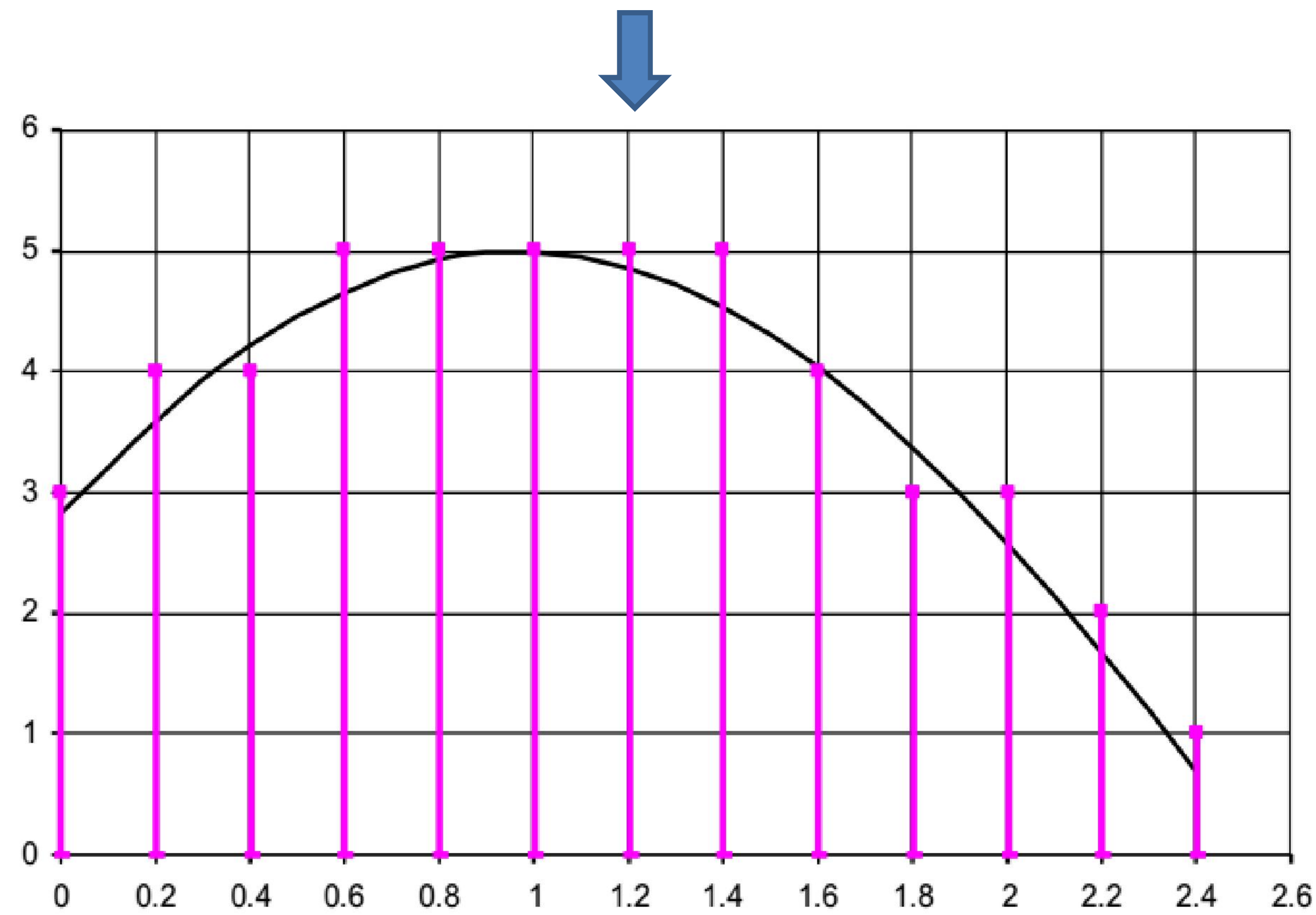
Sampling at the rate of 2*B* at these points will give constant values (zeros)

Signal Reconstruction from Uniform Samples

- Reconstruction objective

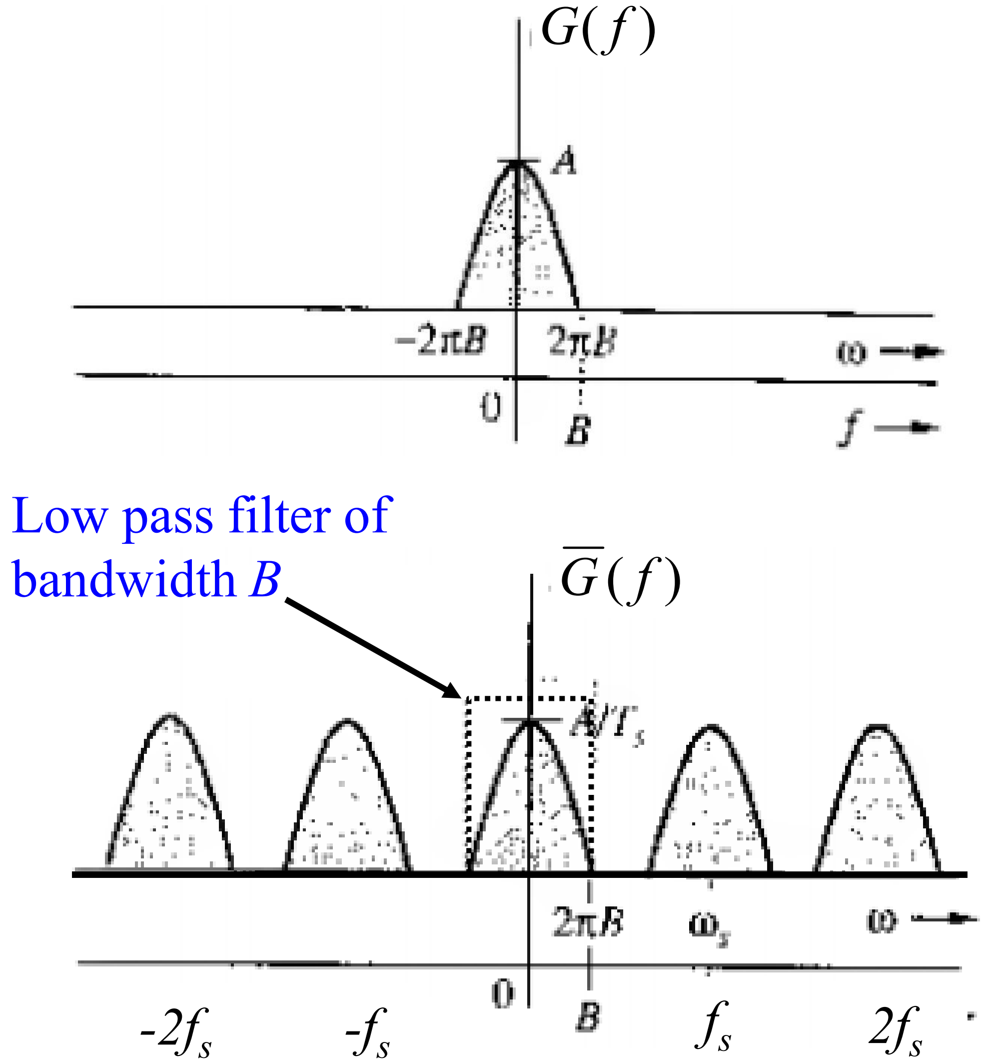
Need to interpolate intermediate values

2.823212	3.58678	4.207355	4.660195	4.927249	......



Signal Reconstruction from Uniform Samples

- We proved theoretically exact reconstruction is possible provide
  - the signal is sampled at rate $f_s > 2B$
  - The sampled signal is passed though an ILPF with bandwidth B



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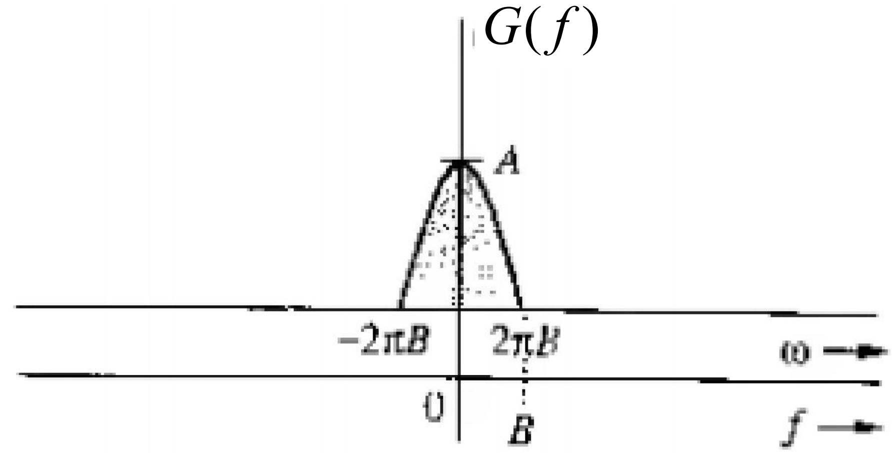
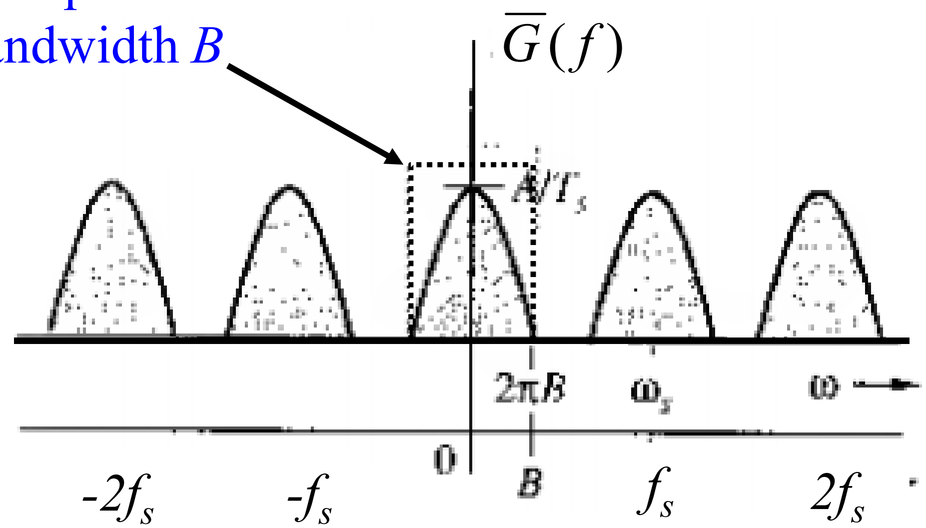


Low pass filter of bandwidth B



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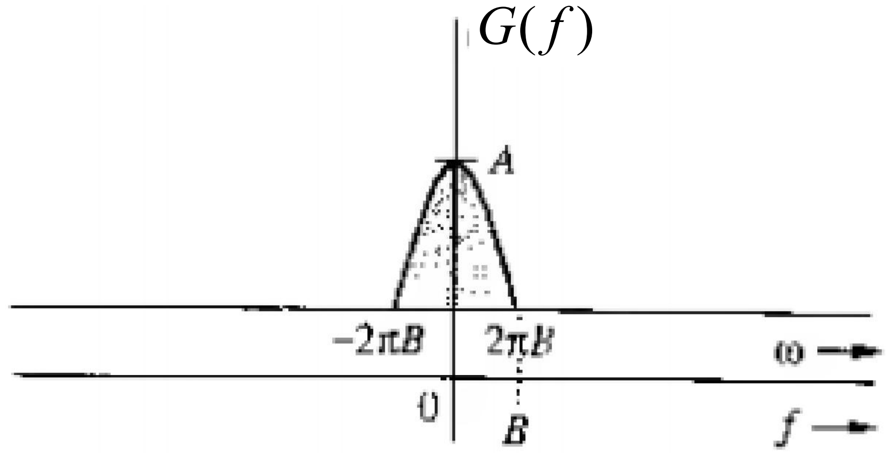

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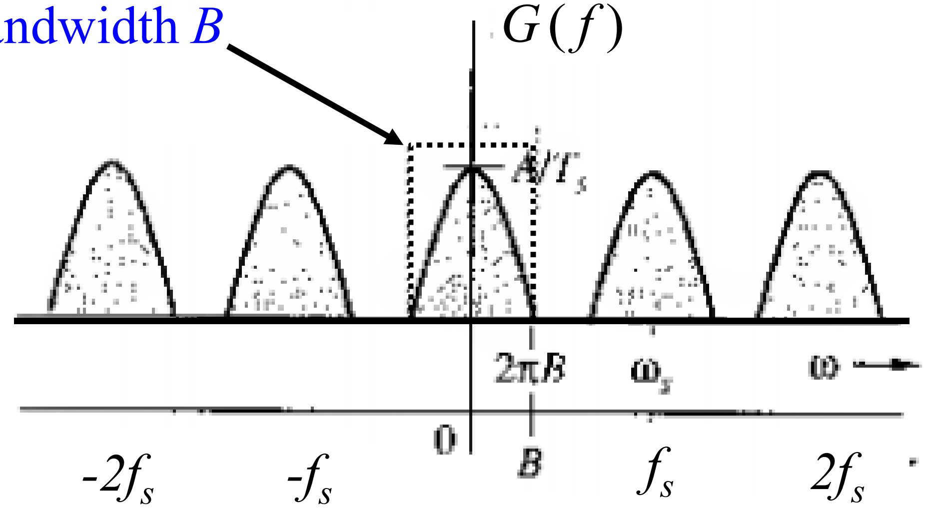
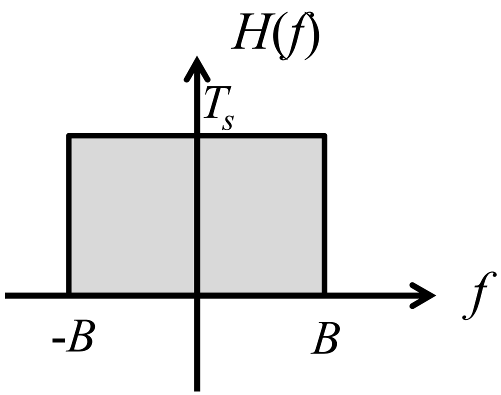


$$\overline{G}(f) = \frac{1}{T_s} \sum_n G(f - nf_s)$$

$$H(f) = T_s \prod\left(\frac{f}{2B}\right)$$

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \prod(\frac{f}{2B})$



$$H(f) = T_s \prod(\frac{f}{2B})$$

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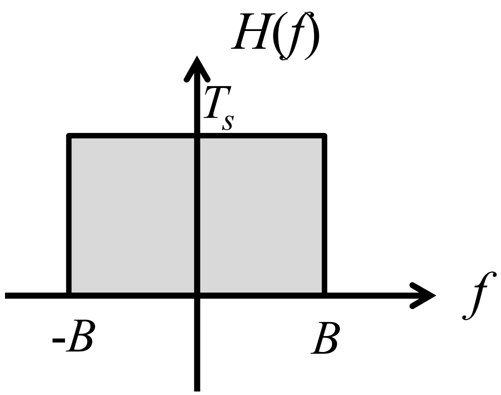

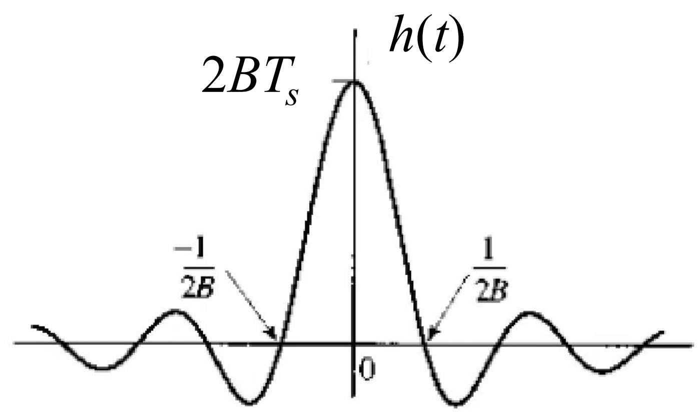

$$H(f) = T_s \prod(\frac{f}{2B})$$

$$h(t) = 2BT_s \operatorname{sinc}(2\pi Bt)$$

Impulse response

Signal Reconstruction from Uniform Samples

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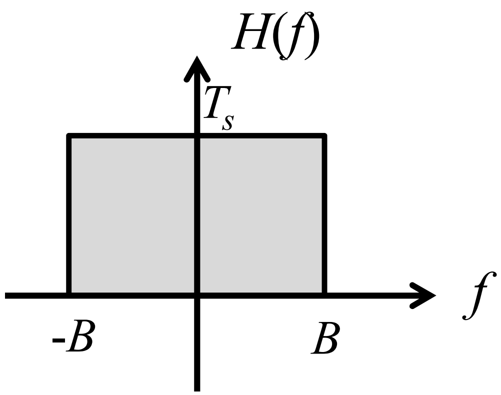
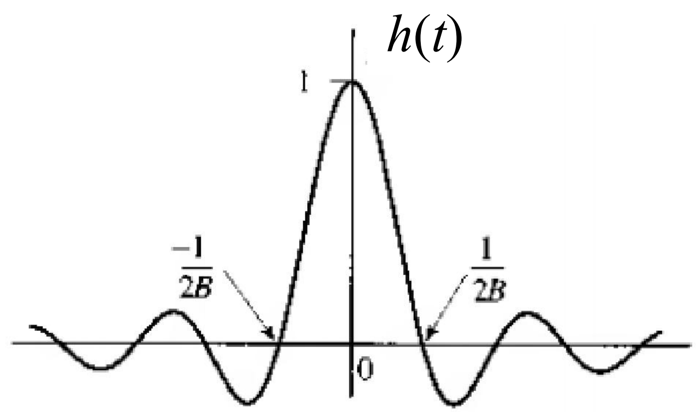




$$H(f) = T_s \prod(\frac{f}{2B})$$

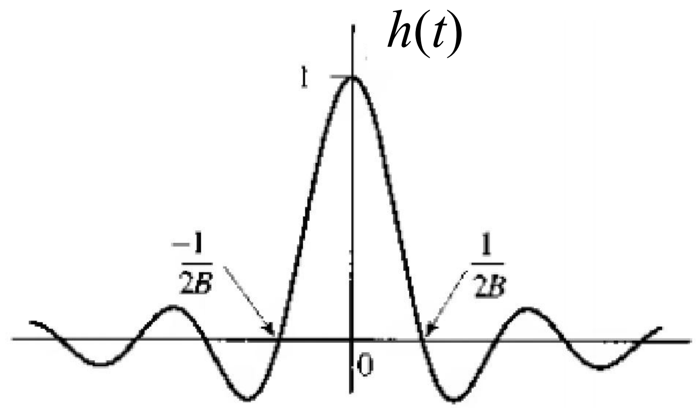
$$h(t) = 2BT_s \mathrm{sinc}(2\pi Bt)$$
$$= \mathrm{sinc}(2\pi Bt)$$

Assuming Nyquist sampling interval $T_s = \frac{1}{2B}$

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \prod(\frac{f}{2B})$

Note: $h(t) = 0$ at Nyquist sampling instant $\pm\frac{n}{2B}$



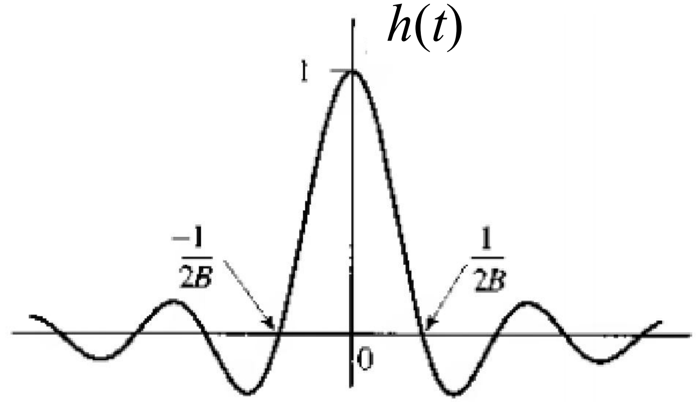
$$h(t) = \text{sinc}(2\pi Bt)$$

Impulse response

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \prod(\frac{f}{2B})$

If we pass $\bar{g}(t)$ through an LTI with this impulse response, the output will be $g(t)$.



$$h(t) = \text{sinc}(2\pi Bt)$$

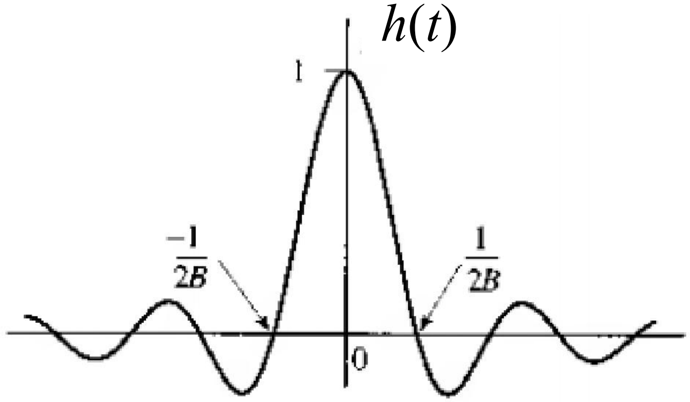
Impulse response

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \prod(\frac{f}{2B})$

If we pass $\overline{g}(t)$ through an LTI with this impulse response, the output will be $g(t)$.

$$\overline{g}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$



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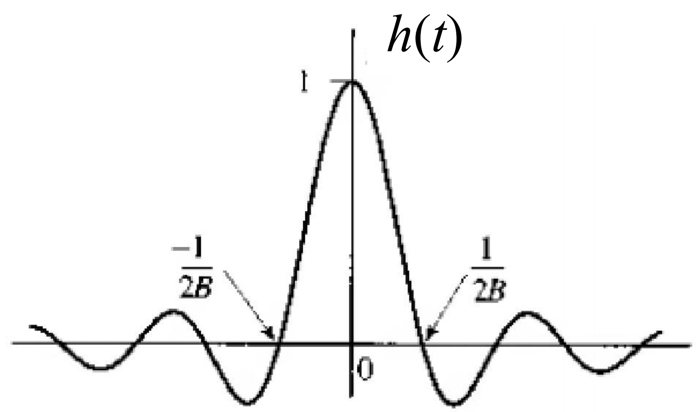

$$h(t) = \text{sinc}(2\pi Bt)$$

$$\sum_k a_k \delta(t - t_k) \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow \sum_k a_k h(t - t_k)$$

Signal Reconstruction from Uniform Samples

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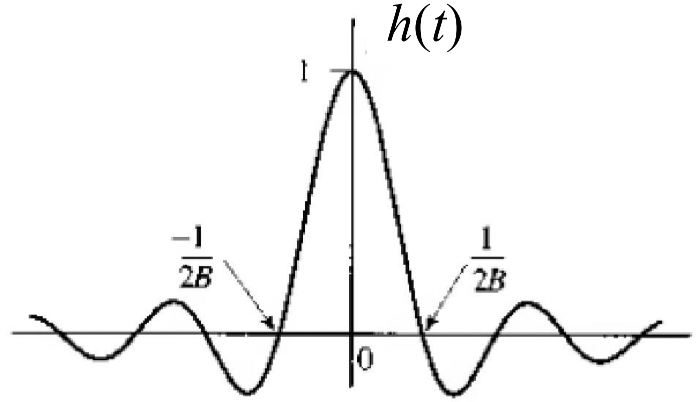


$$h(t) = \text{sinc}(2\pi Bt)$$

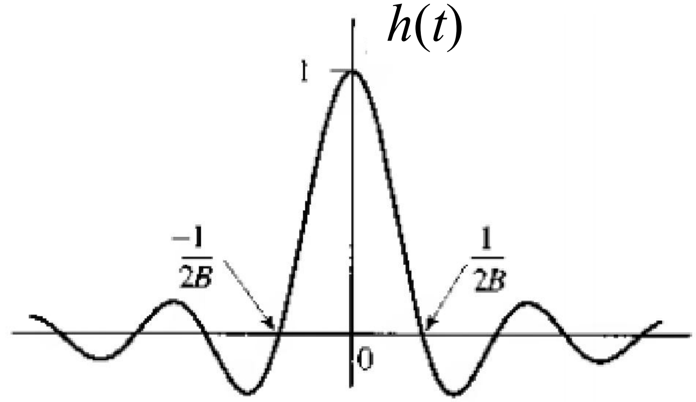
$$x(t) = \sum_n \delta(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\end{array}} \longrightarrow y(t) = \sum_n h(t - \nabla\tau)x(n\nabla\tau)\nabla\tau$$

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \prod(\frac{f}{2B})$

If we pass $\bar{g}(t)$ through an LTI with this impulse response, the output will be $g(t)$.

$$\bar{g}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$



$$h(t) = \text{sinc}(2\pi Bt)$$

$$\bar{g}(t) = \sum_n g(nT_s)\delta(t - nT_s) \longrightarrow \boxed{\begin{array}{c}\text{Linear Time} \\ \text{Invariant System,} \\ h(t)\end{array}} \longrightarrow g(t) = \sum_n g(nT_s)h(t - nT_s)$$

Signal Reconstruction from Uniform Samples

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$$= \bar{g}(t) * h(t)$$

Signal Reconstruction from Uniform Samples

$$\bar{g}(t) = \sum_n g(nT_s)\delta(t - nT_s) \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\\ = \mathrm{sinc}(2\pi Bt)\end{array}} \longrightarrow g(t) = \sum_n g(nT_s)h(t - nT_s)$$

$$= \bar{g}(t) * h(t)$$

$$\begin{aligned} g(t) &= \sum_n g(nT_s)h(t - nT_s) \\ &= \sum_n g(nT_s)\mathrm{sinc}\left[2\pi B(t - nT_s)\right] \\ &= \sum_n g(nT_s)\mathrm{sinc}(2\pi Bt - n\pi) \end{aligned}$$

Signal Reconstruction from Uniform Samples

$$\bar{g}(t)=\sum_n g(nT_s)\delta(t-nT_s) \longrightarrow \boxed{\begin{array}{c}\text{Linear Time}\\ \text{Invariant System,}\\ h(t)\\ =\mathrm{sinc}(2\pi Bt)\end{array}} \longrightarrow g(t)=\sum_n g(nT_s)h(t-nT_s)$$

$$=\bar{g}(t)*h(t)$$

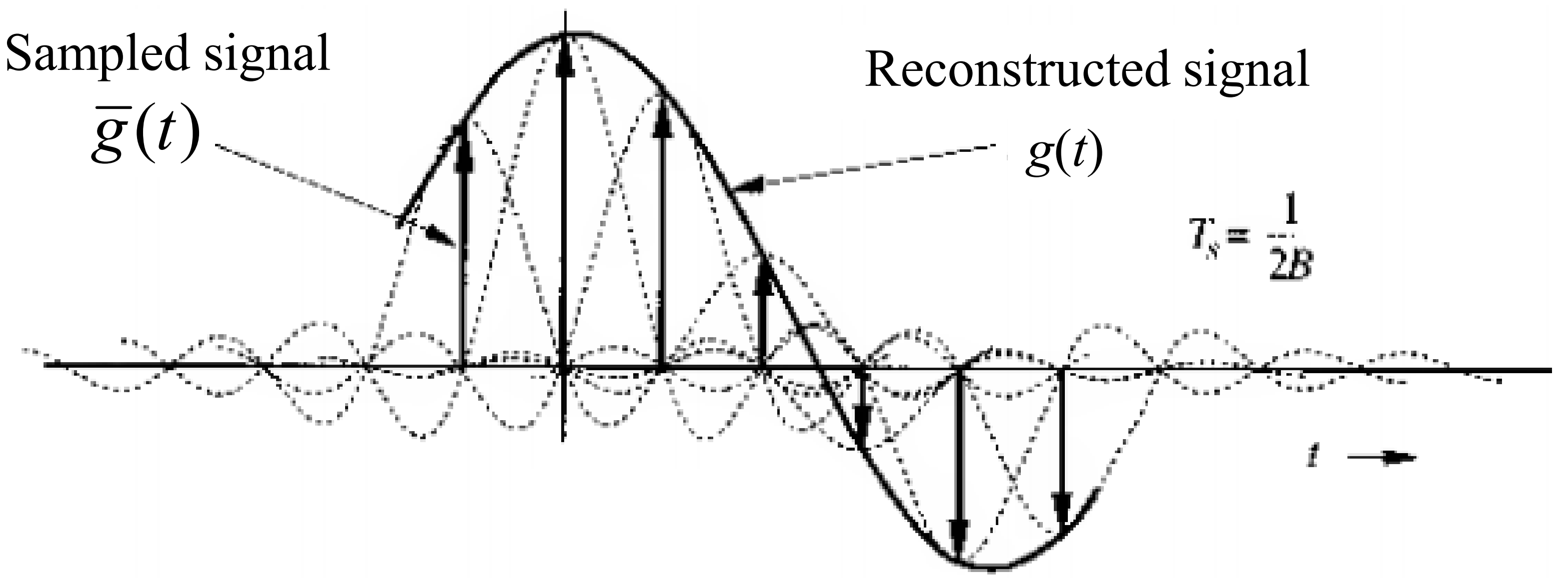
$$g(t)=\sum_n g(nT_s)\,\mathrm{sinc}(2\pi Bt-n\pi)$$

Interpolation formula to find the value $g(t)$ at arbitrary time t.

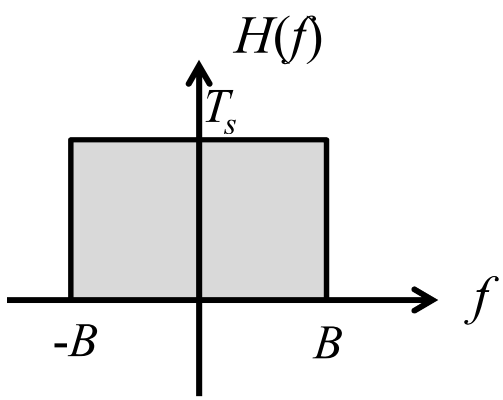
Signal Reconstruction from Uniform Samples

Significance of Interpolation formula

$$g(t) = \sum_n g(nT_s)\text{sinc}(2\pi Bt - n\pi)$$



Practical Signal Reconstruction (Interpolation)



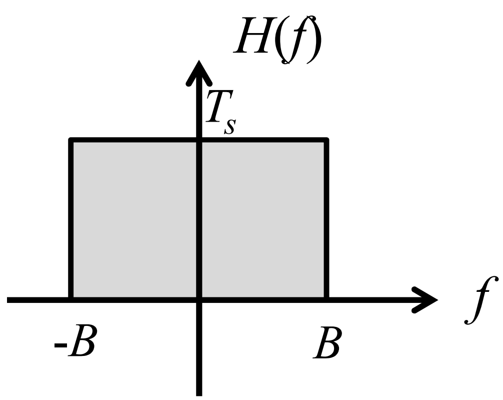
ILPF

generates

$$g(t) = \sum_n g(nT_s)\,\mathrm{sinc}(2\pi Bt - n\pi)$$

Interpolation formula

Practical Signal Reconstruction (Interpolation)



ILPF:
Noncausal
and
Unrealizable

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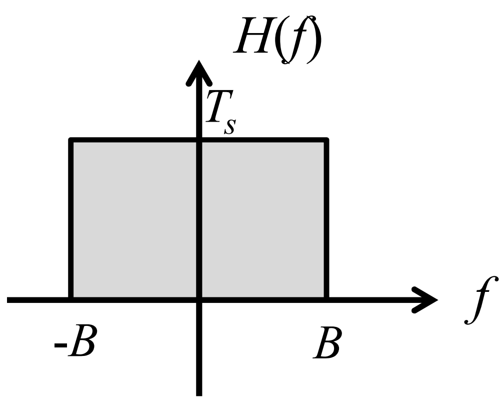
$$g(t) = \sum_n g(nT_s)\,\mathrm{sinc}(2\pi Bt - n\pi)$$

Interpolation formula

Practical Signal Reconstruction (Interpolation)



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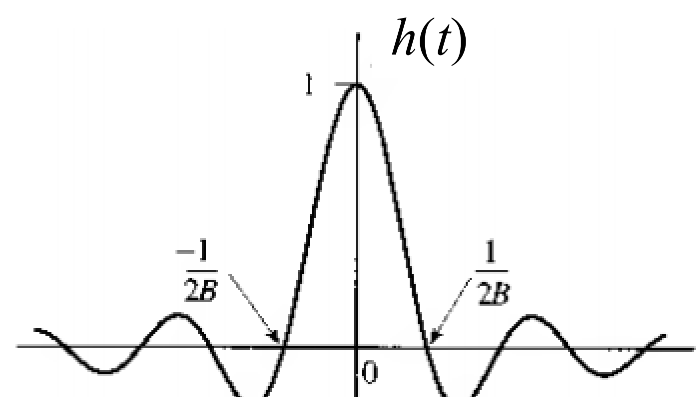
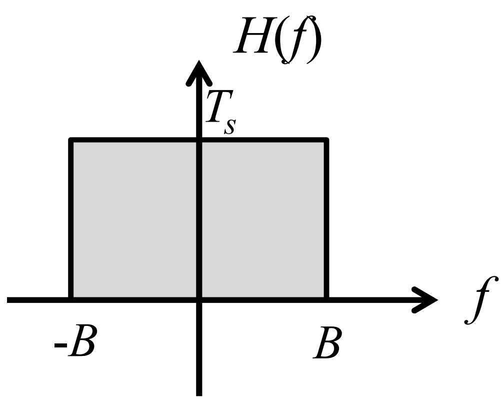


Also seen form long nature of sinc reconstruction pulse

Practical Signal Reconstruction (Interpolation)

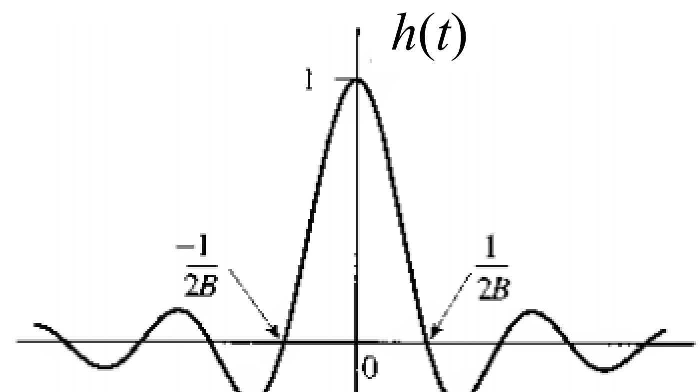


generates

$$g(t) = \sum_n g(nT_s)\,\mathrm{sinc}(2\pi Bt - n\pi)$$

Interpolation formula

Sharp transition of ILPF at $\pm B$ causes this problem



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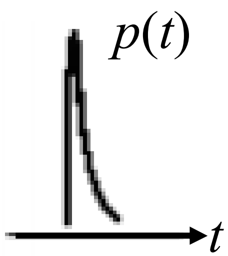
Practical Signal Reconstruction (Interpolation)

Alternate solution is to use realizable pulse for interpolation



Practical Signal Reconstruction (Interpolation)

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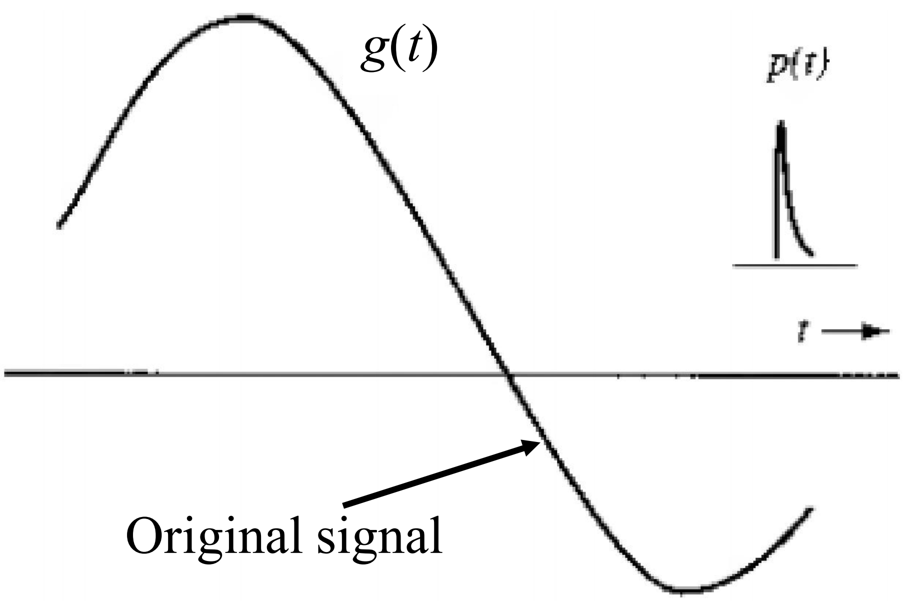

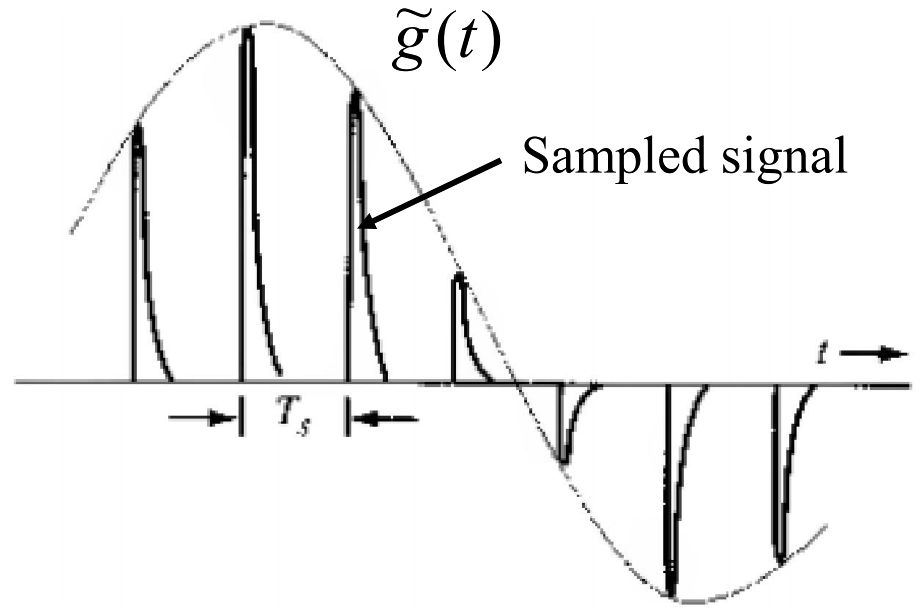

Practical Signal Reconstruction (Interpolation)

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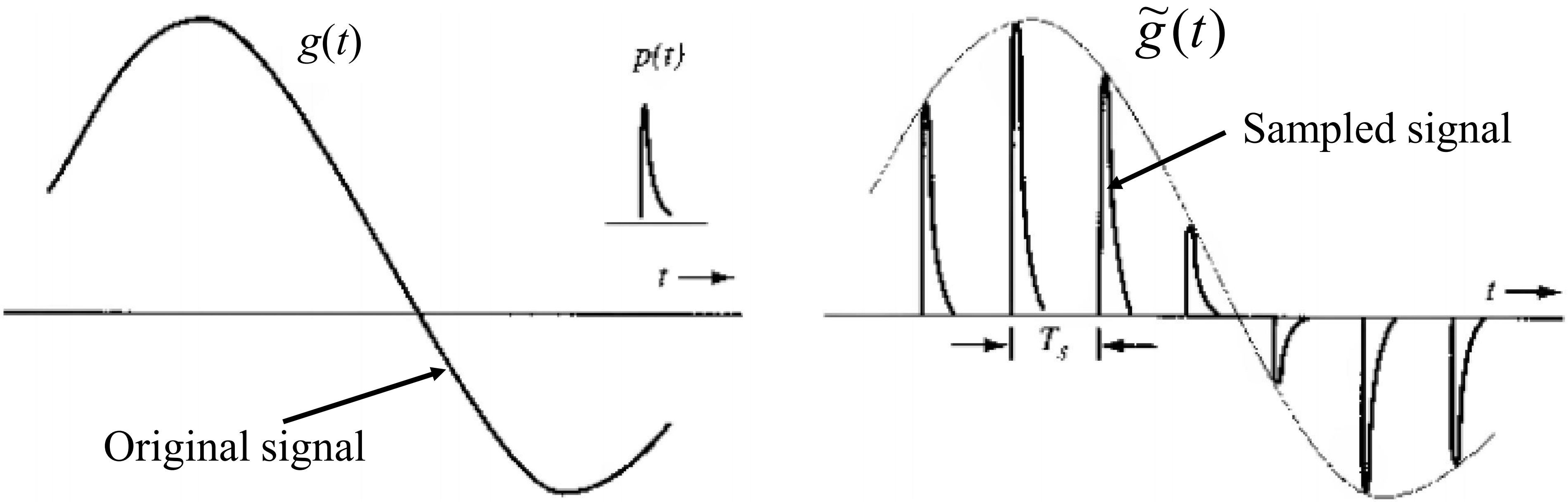
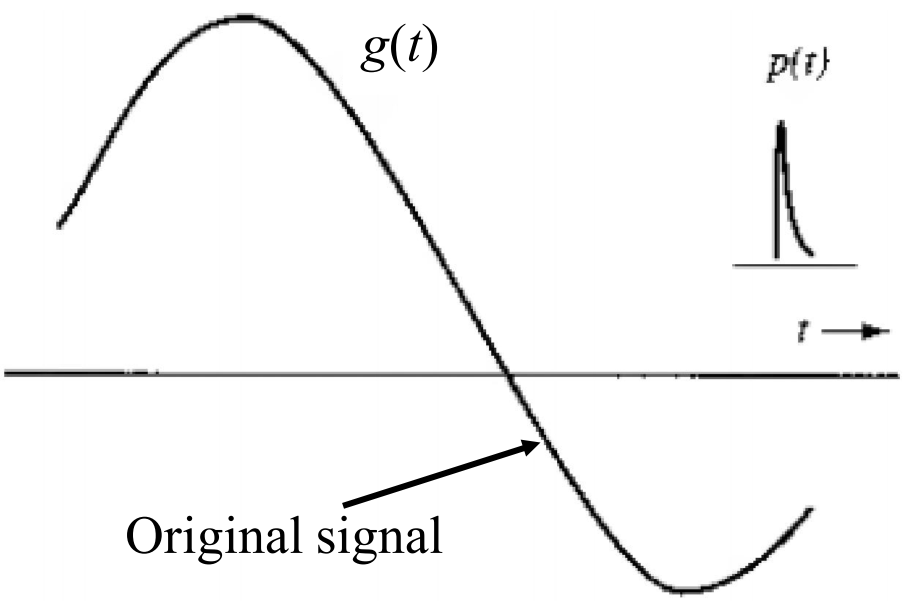
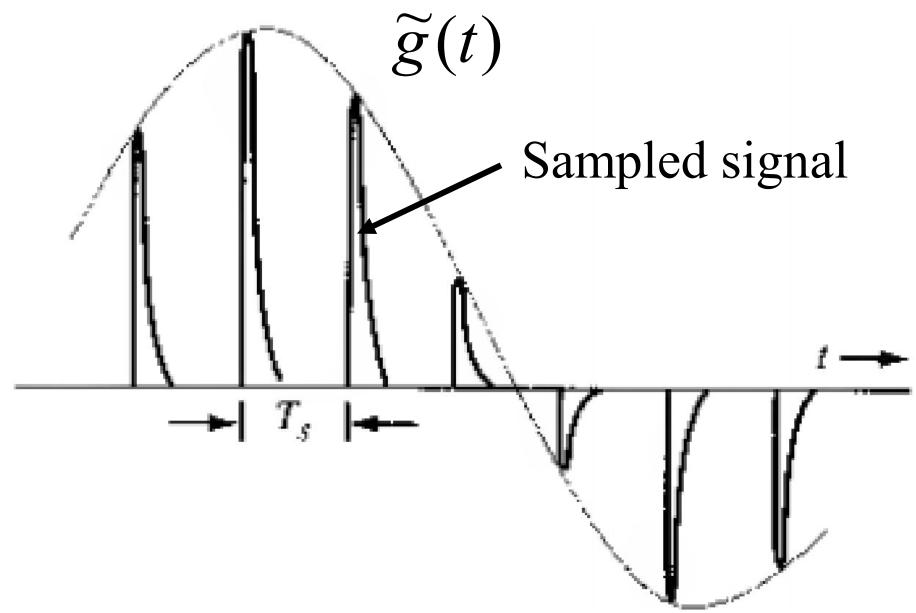


$$\tilde{g}(t) = \sum_n g(nT_s)\,p(t - nT_s)$$

Practical Signal Reconstruction (Interpolation)

Alternate solution is to use realizable pulse for interpolation





$$\tilde{g}(t) = \sum_n g(nT_s)\,p(t - nT_s) = \bar{g}(t) * p(t)$$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

$$\tilde{g}(t) = \sum_n g(nT_s)\, p(t - nT_s) = \bar{g}(t) * p(t)$$

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In frequency domain, it becomes

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

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Remember, $\bar{G}(f) = \frac{1}{T_s}\sum_n G(f - nf_s)$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

$$\tilde{g}(t) = \sum_n g(nT_s)\, p(t - nT_s) = \bar{g}(t) * p(t)$$

In frequency domain, it becomes

$$\tilde{G}(f) = P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$

The reconstructed signal $\tilde{g}(t)$ also contains multiples replicas of $G(f)$ at every f_s distance apart and filtered by $P(f)$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

$$\tilde{g}(t) = \sum_{n} g(nT_s)\, p(t - nT_s) = \bar{g}(t) * p(t)$$

In frequency domain, it becomes

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_{n} G(f - nf_s)$$

To fully recover $g(t)$ from $\tilde{g}(t)$, we need another filter or Equalizer $E(f)$

Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

To fully recover $g(t)$ from $\tilde{g}(t)$, we need another filter or Equalizer $E(f)$

That means,

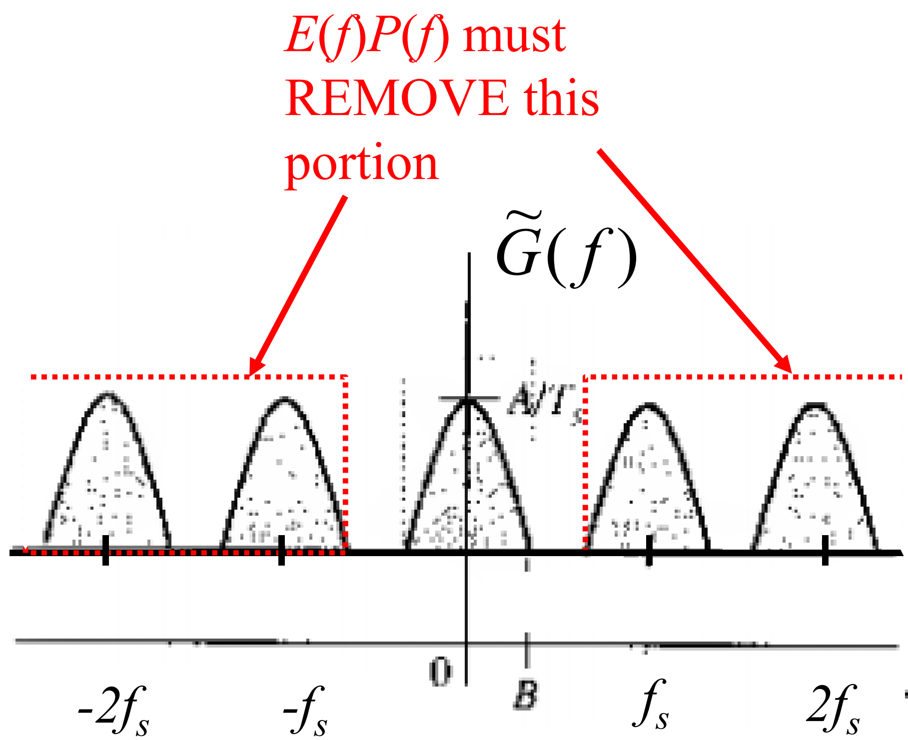
$$\begin{aligned} G(f) &= E(f)\tilde{G}(f) \\ &= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s) \end{aligned}$$

Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

$$G(f) = E(f)\tilde{G}(f)$$

$$= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$



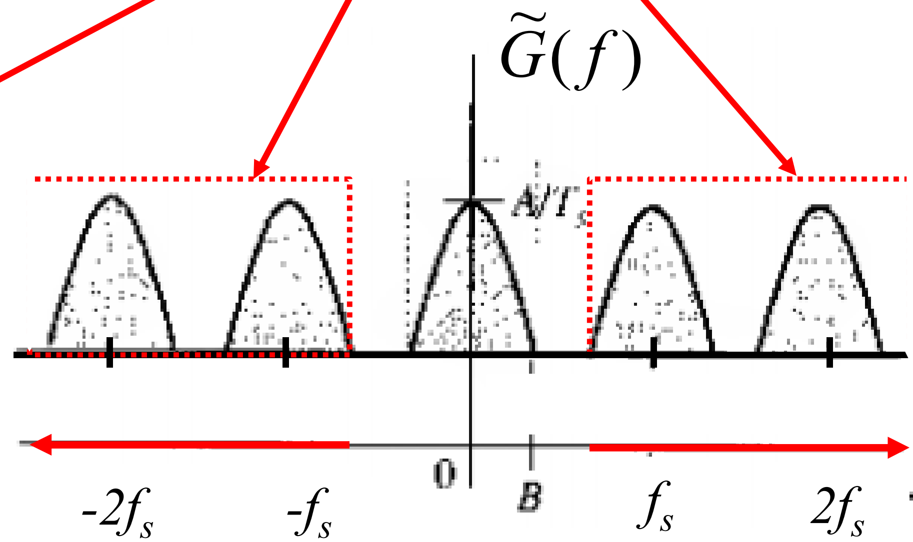
Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

$$G(f) = E(f)\tilde{G}(f)$$
$$= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

$E(f)P(f)$ must REMOVE this portion

$$\boxed{E(f)P(f) = 0 \quad |f| > f_s - B}$$



Practical Signal Reconstruction (Interpolation)

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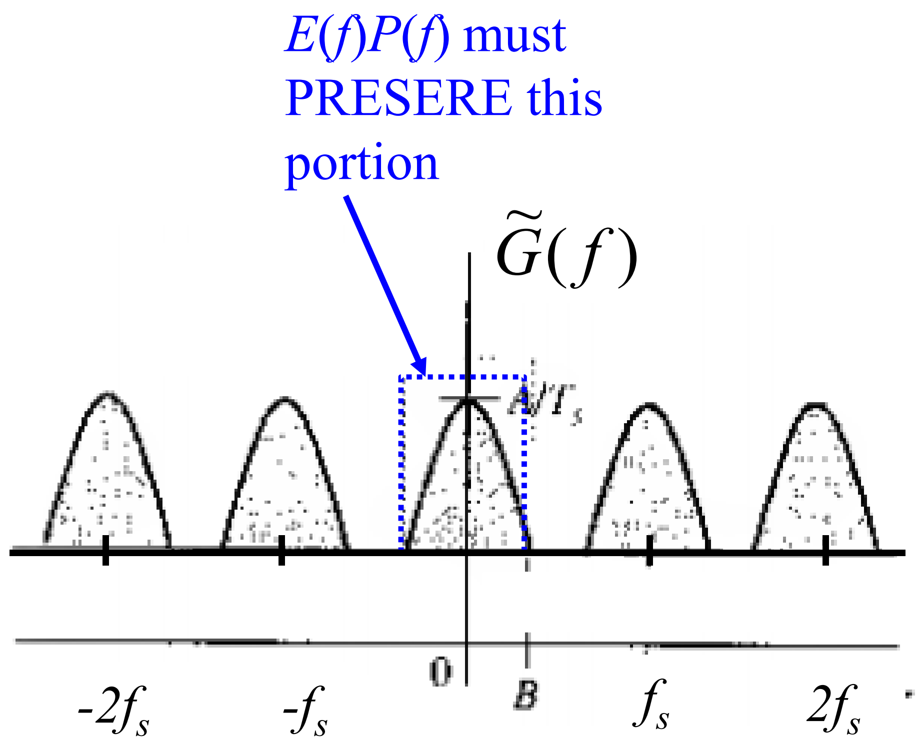


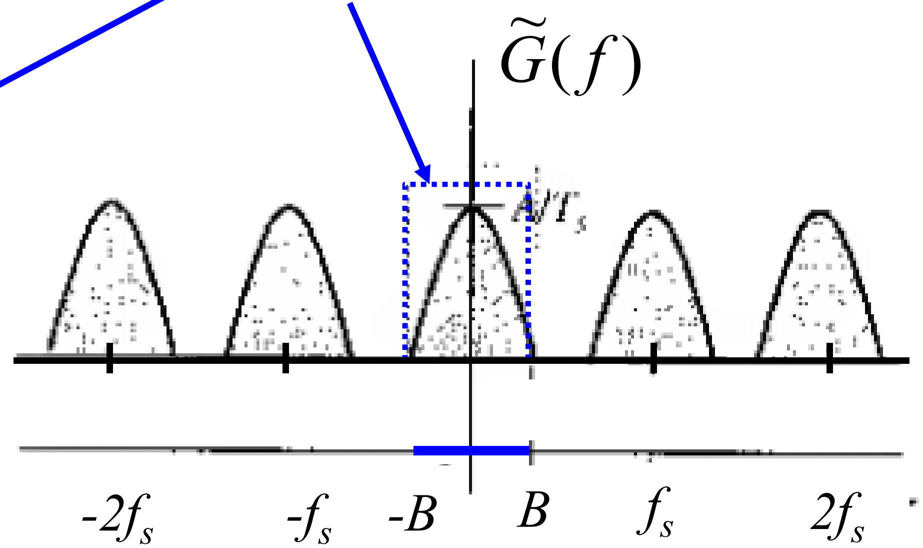
Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

$$G(f) = E(f)\tilde{G}(f)$$
$$= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

$$\boxed{E(f)P(f) = T_s \quad |f| < B}$$

$E(f)P(f)$ must PRESERE this portion



Practical Signal Reconstruction (Interpolation)

$$\widetilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

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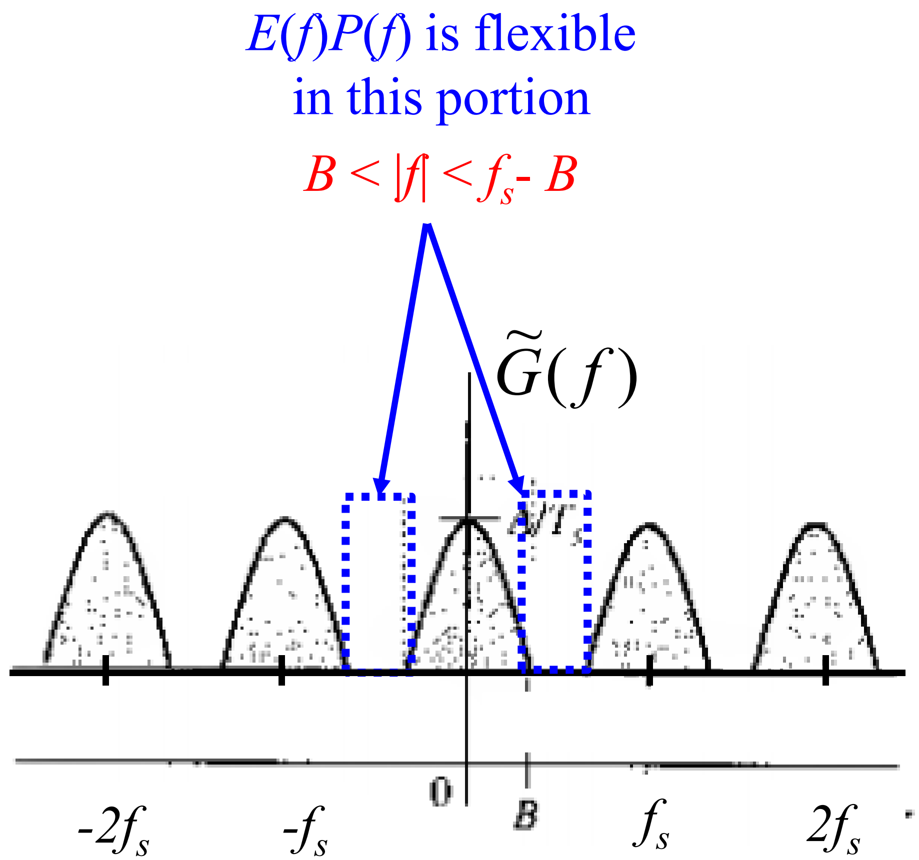
$$= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$



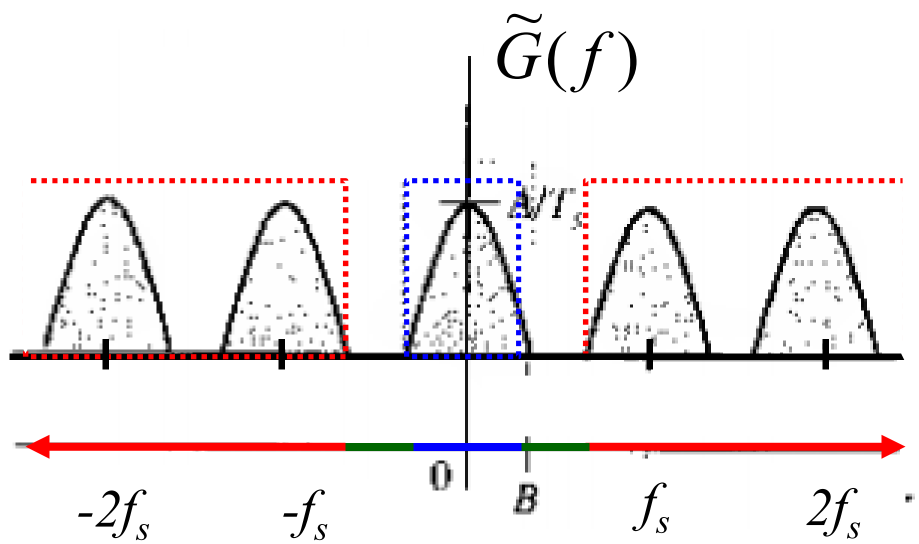
Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

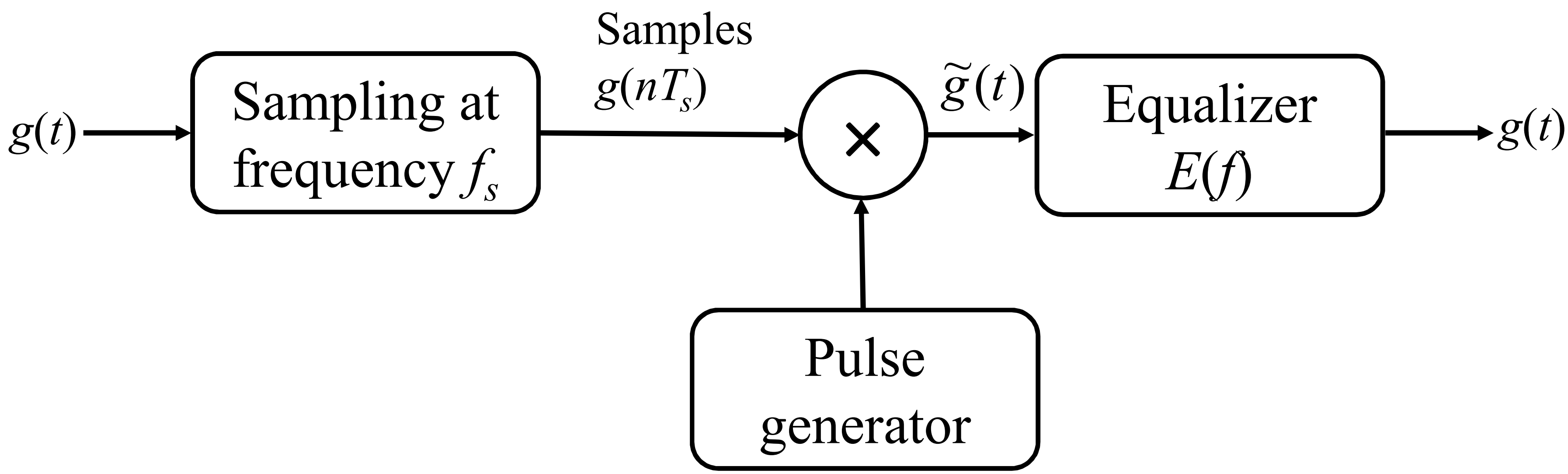
$$G(f) = E(f)\tilde{G}(f)$$
$$= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

3 conditions together

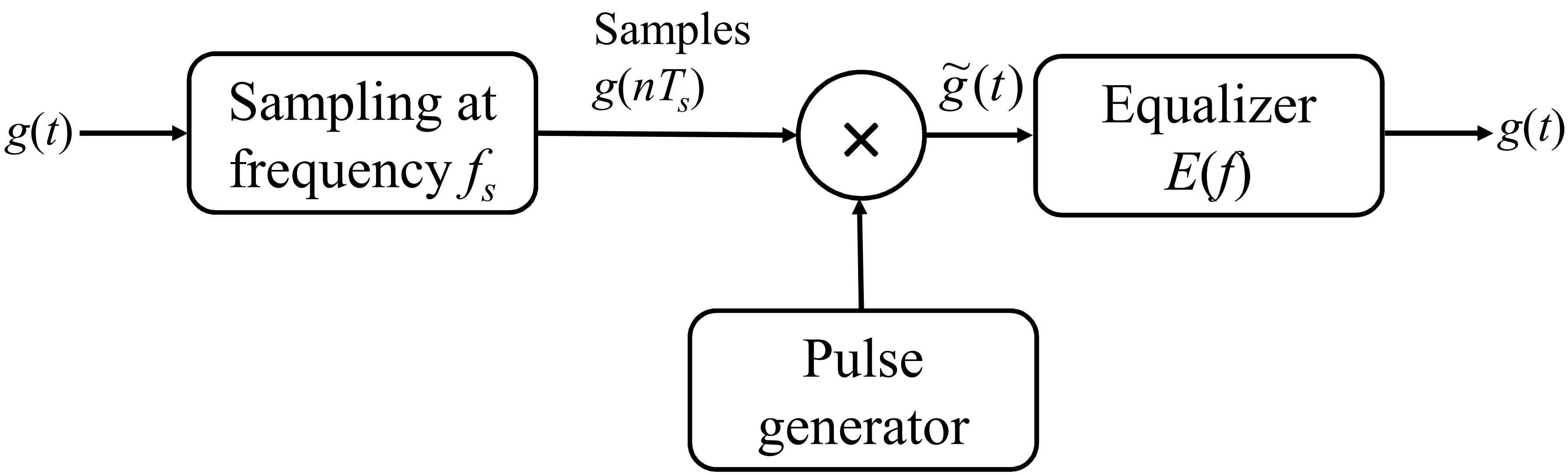
$$E(f) = \begin{cases} T_s / P(f) & |f| \le B \\ \text{Flexible} & B < |f| \le f_s - B \\ 0 & |f| \ge f_s - B \end{cases}$$



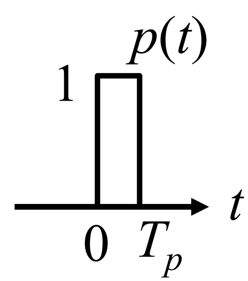
Practical Signal Reconstruction (Interpolation)



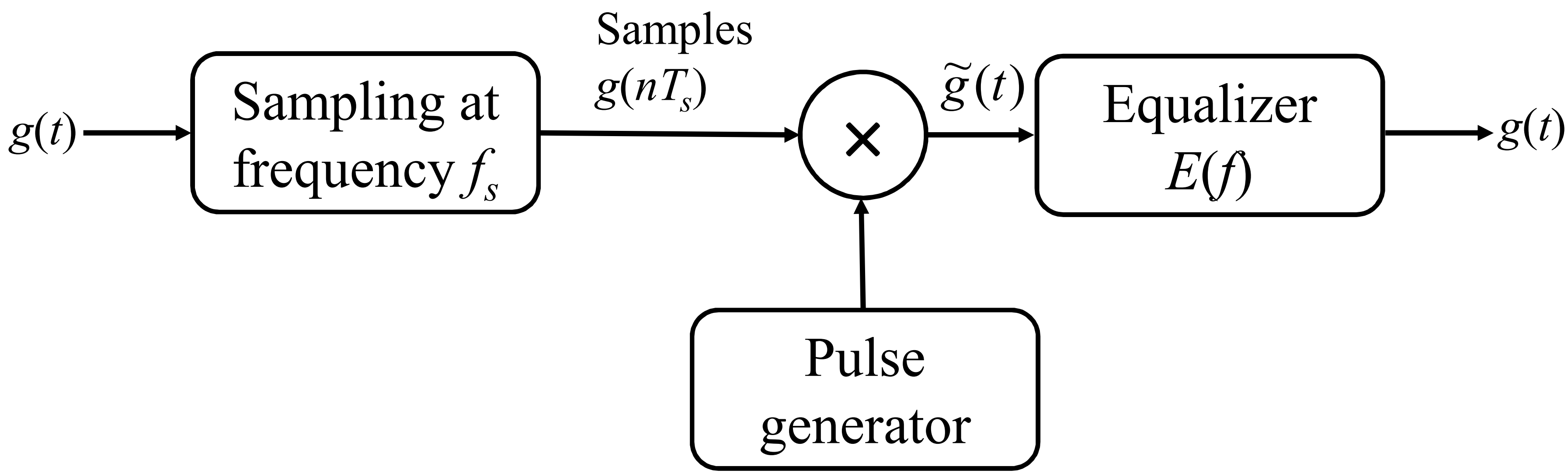
Practical Signal Reconstruction (Interpolation)



Let this pulse generator generates this short pulse



Practical Signal Reconstruction (Interpolation)



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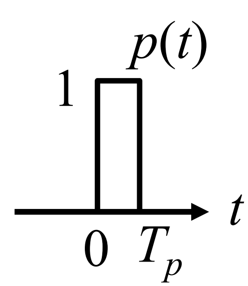
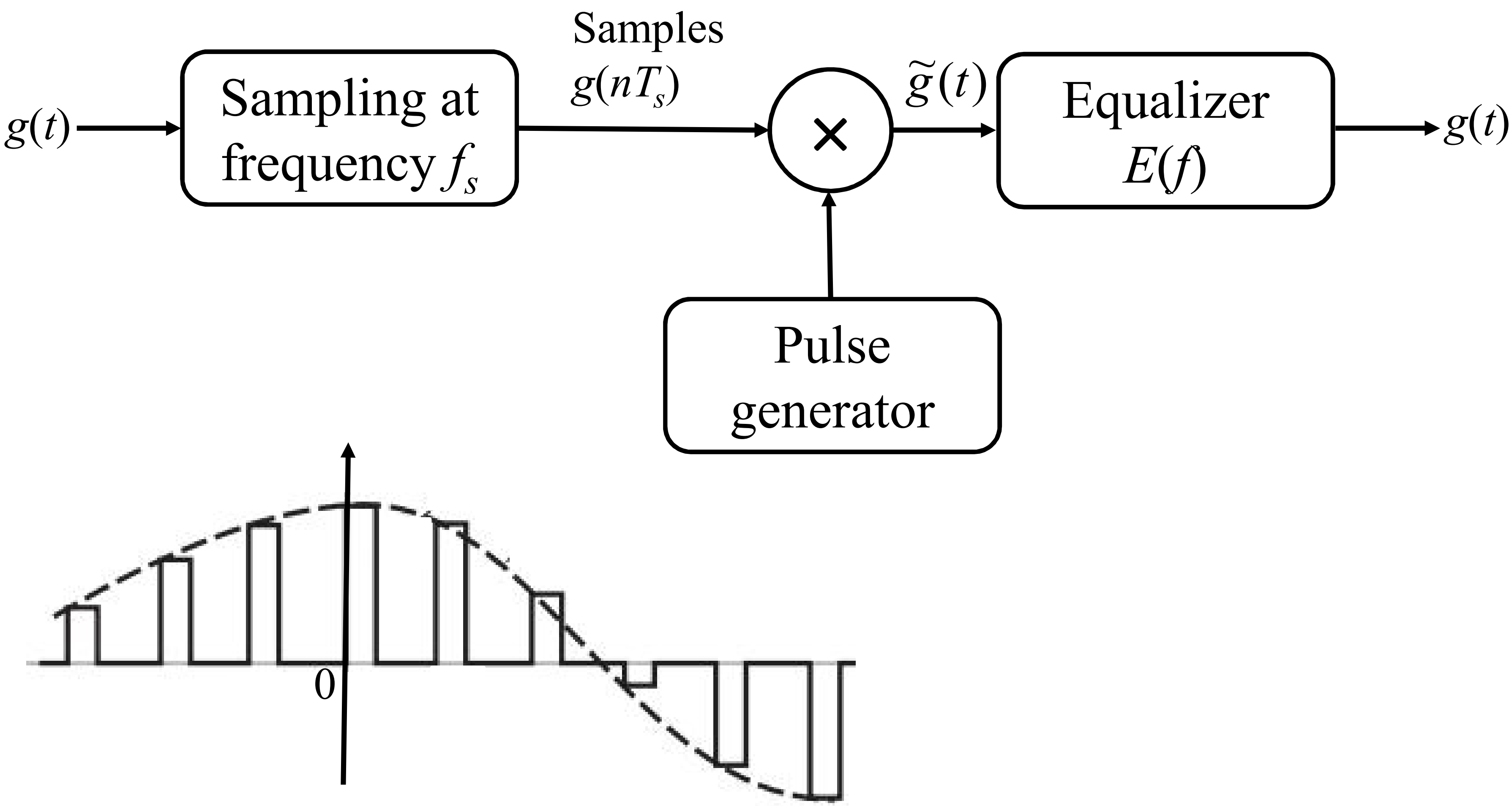
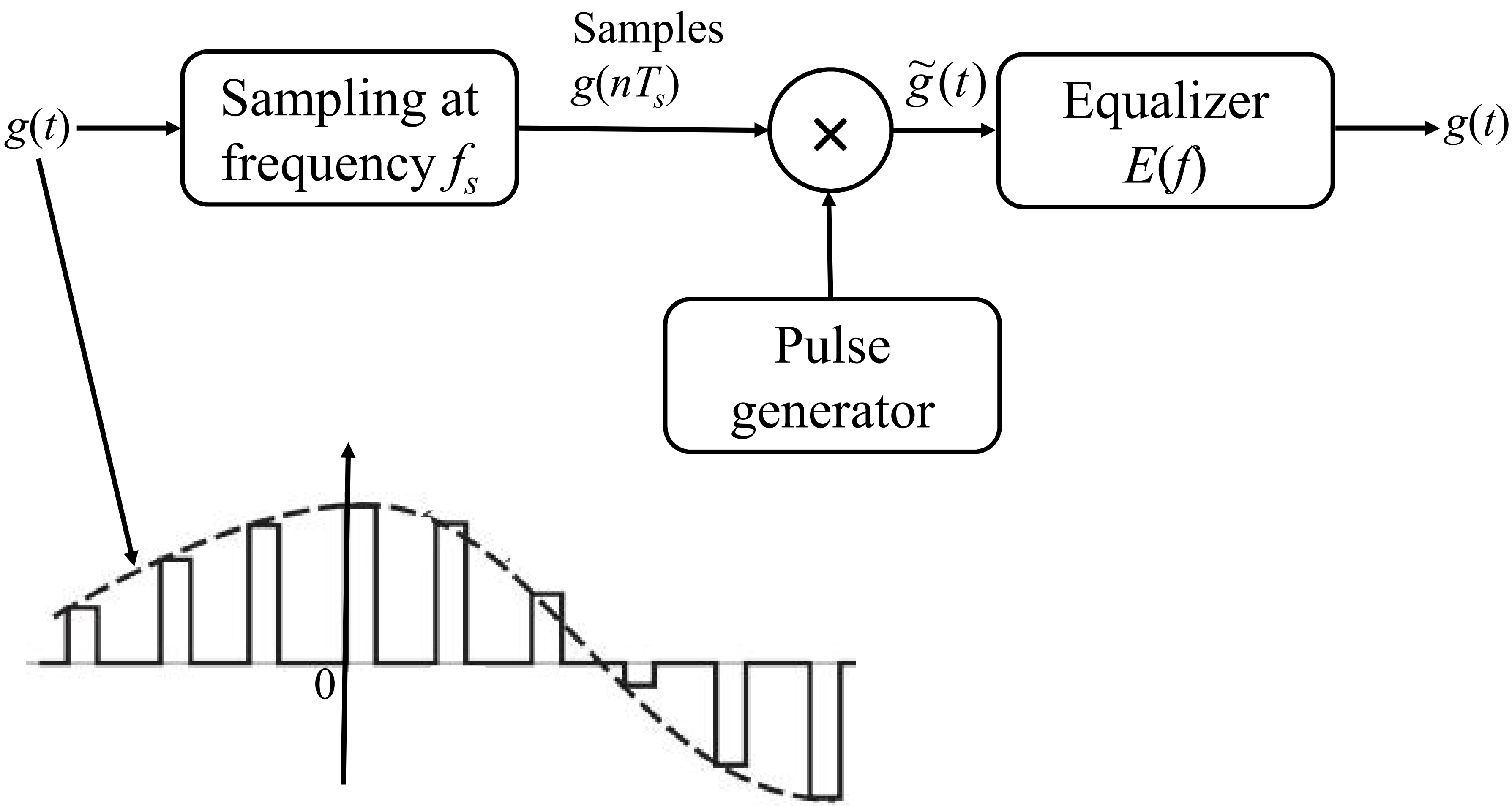


$$p(t) = \Pi\left(\frac{t - 0.5T_p}{T_p}\right)$$

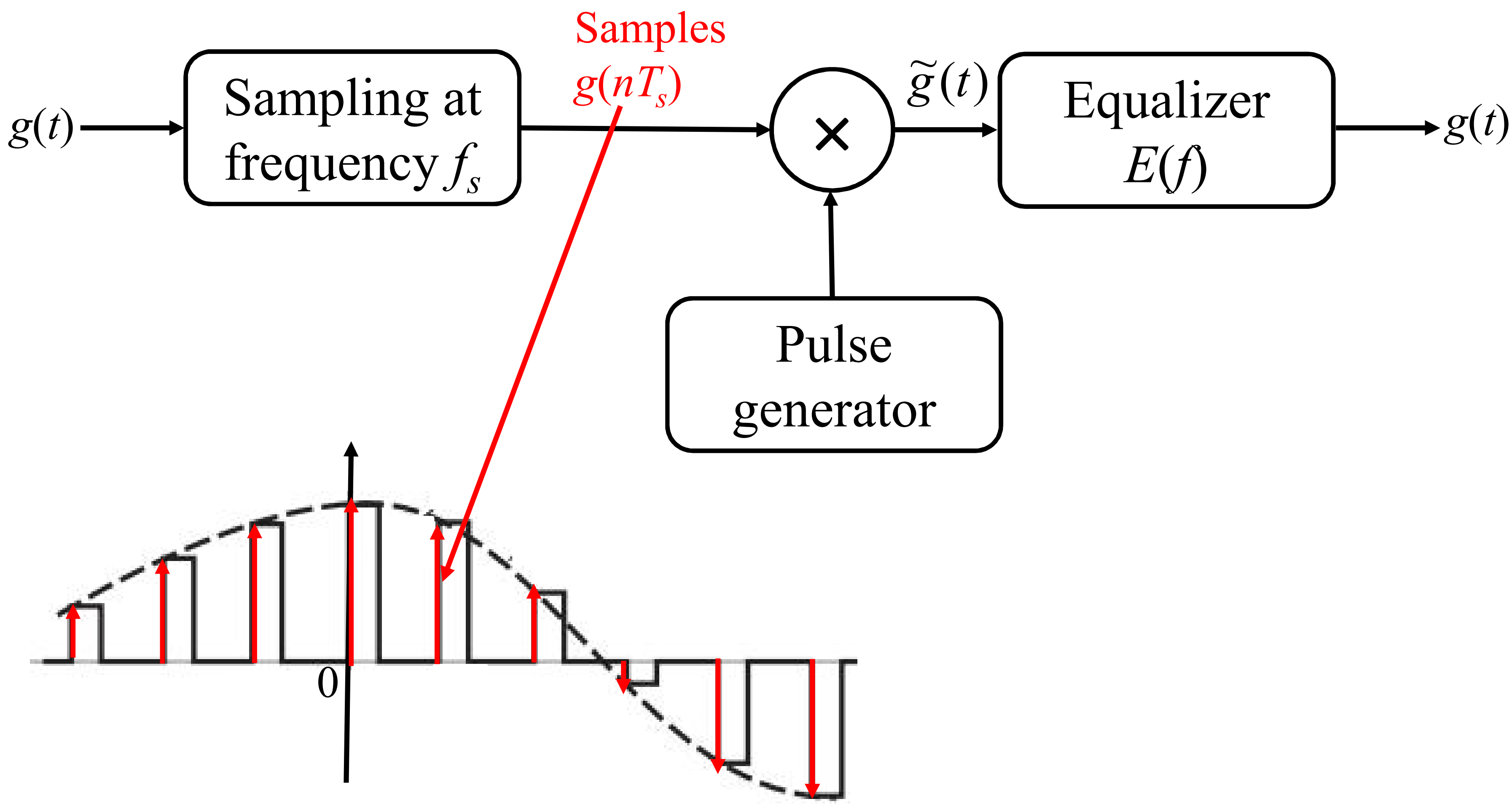
Practical Signal Reconstruction (Interpolation)



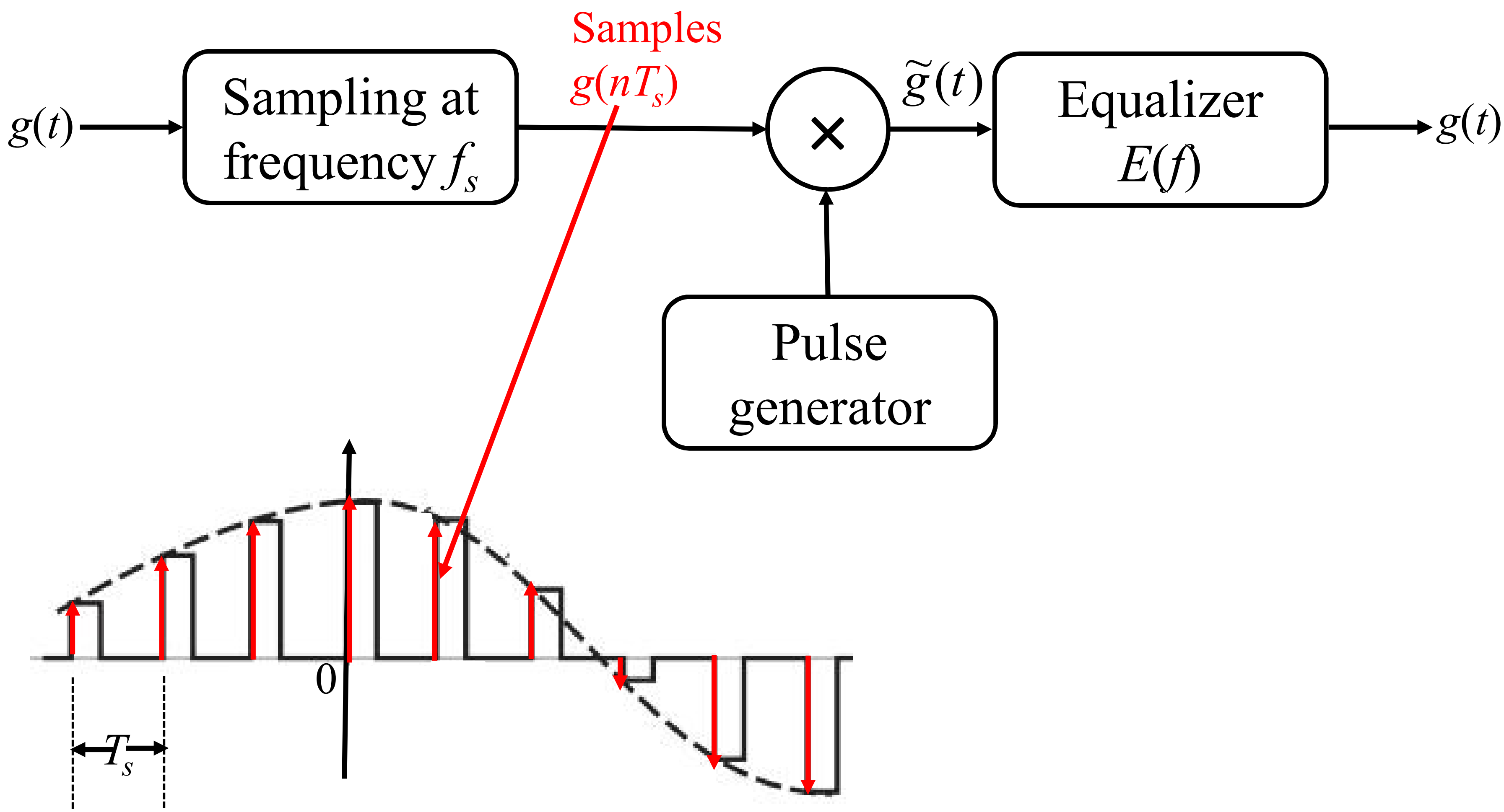
Practical Signal Reconstruction (Interpolation)



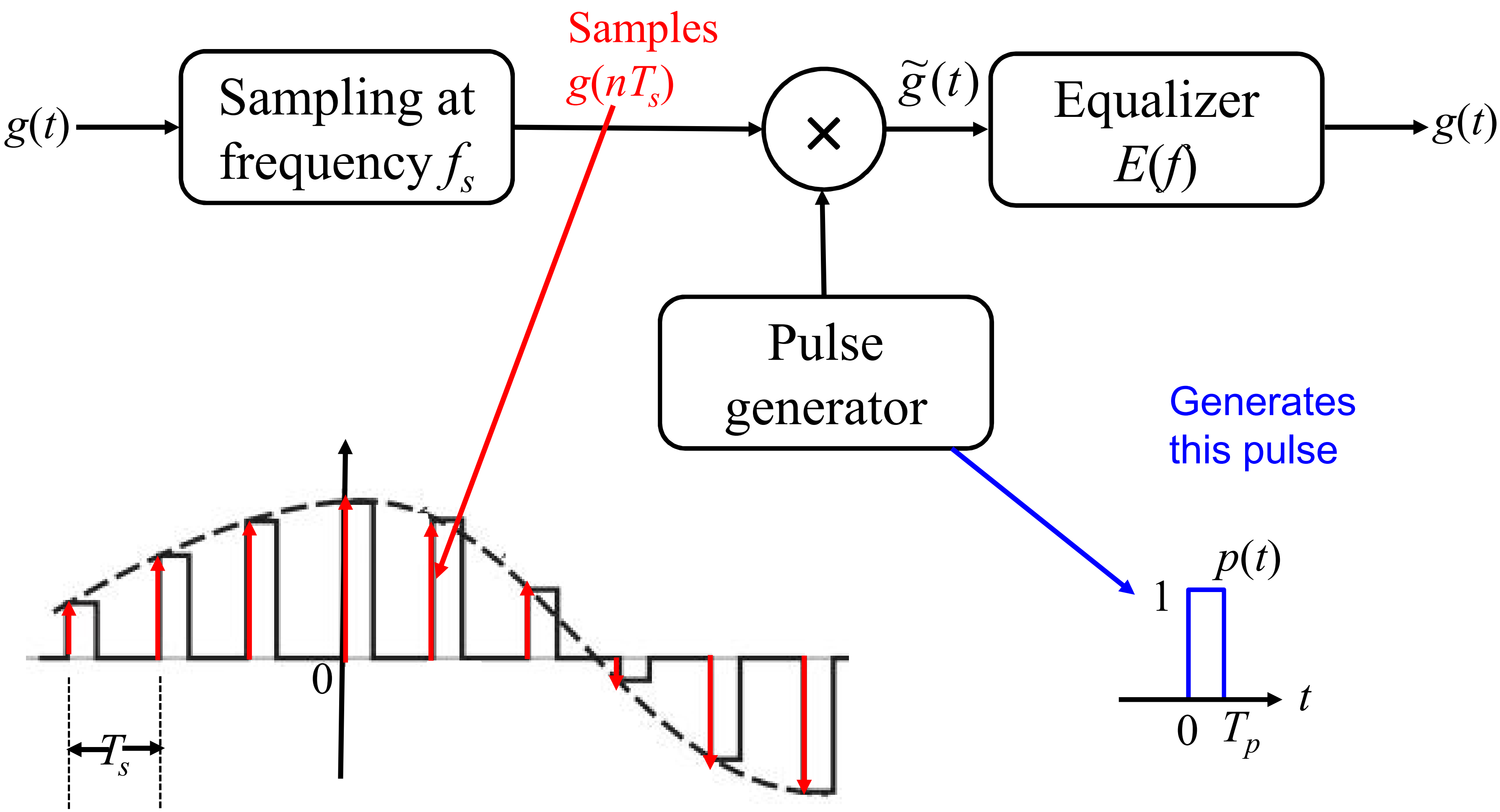
Practical Signal Reconstruction (Interpolation)



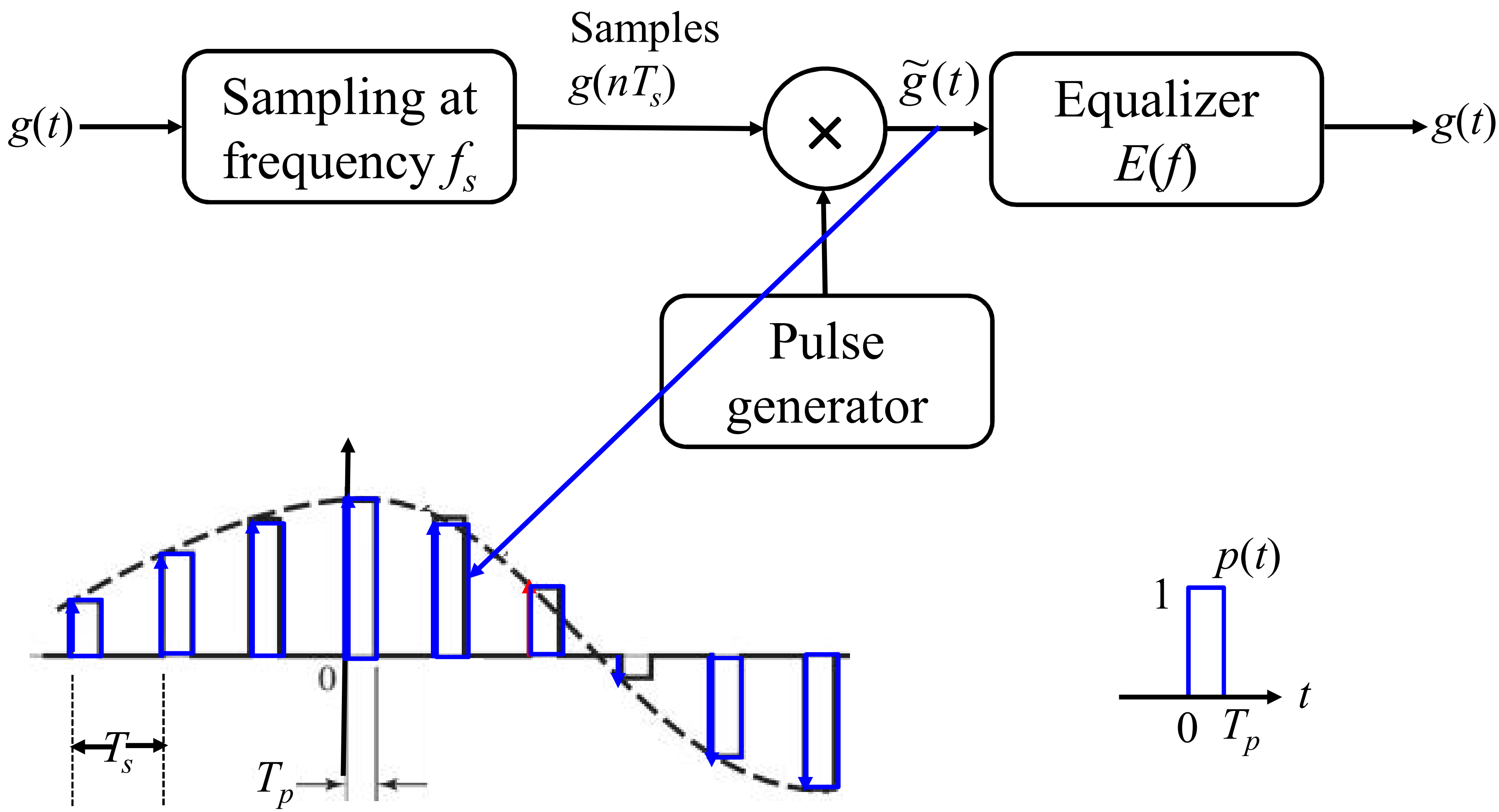
Practical Signal Reconstruction (Interpolation)



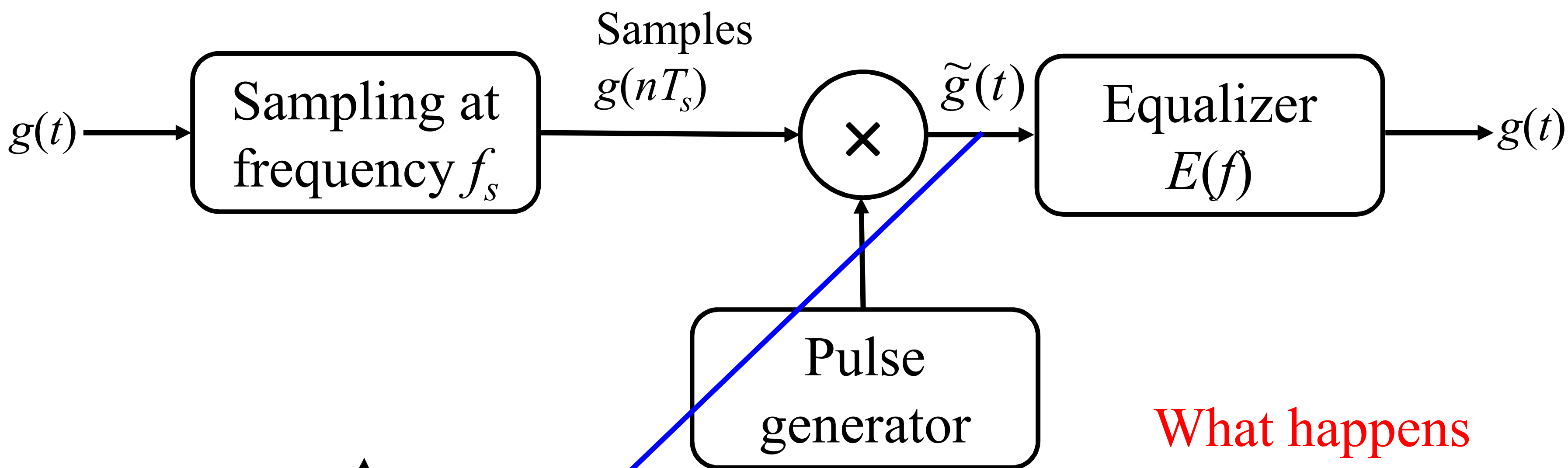
Practical Signal Reconstruction (Interpolation)

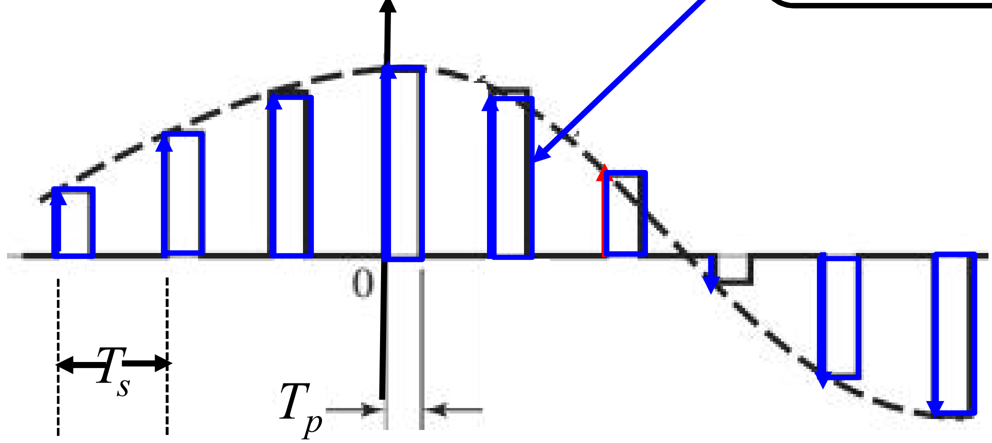


Practical Signal Reconstruction (Interpolation)

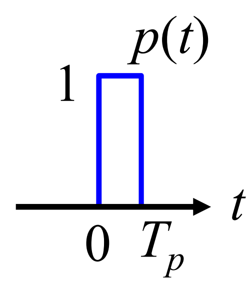


Practical Signal Reconstruction (Interpolation)

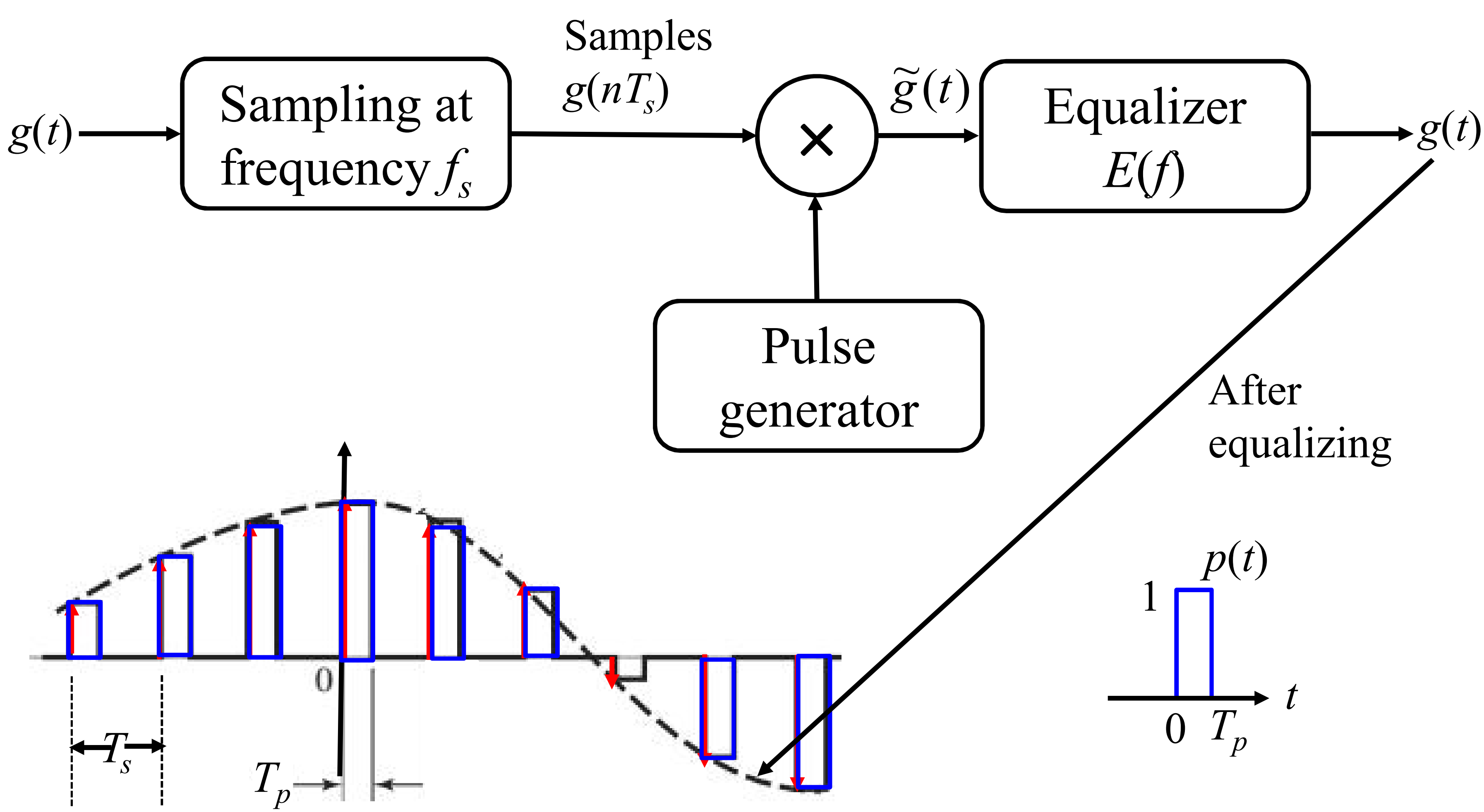




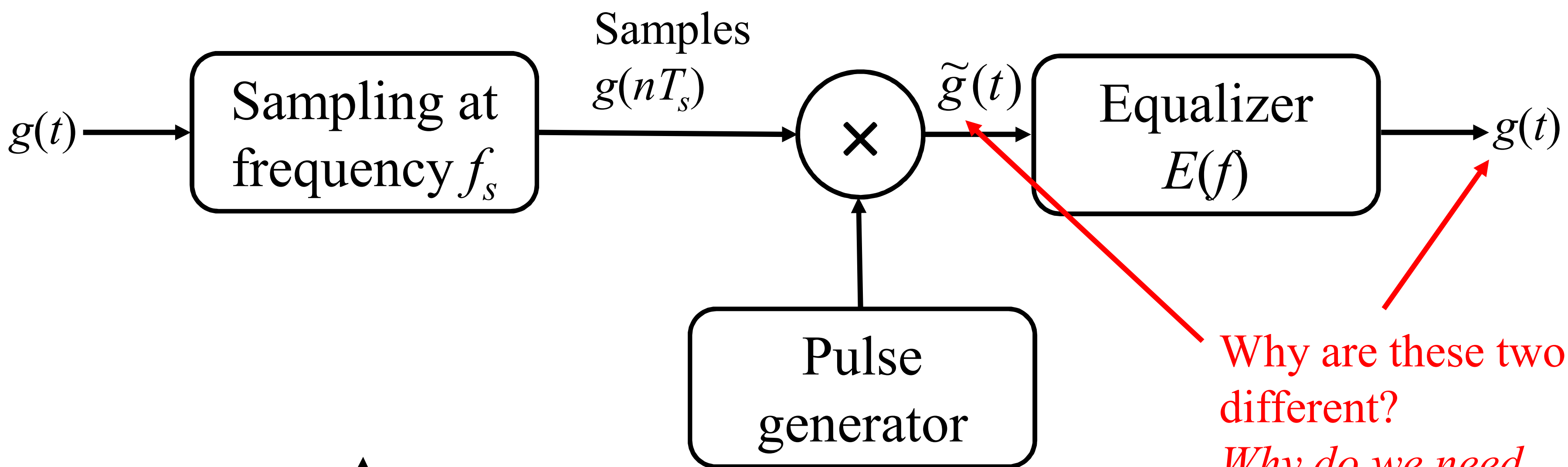
What happens if $T_p \to 0$?



Practical Signal Reconstruction (Interpolation)

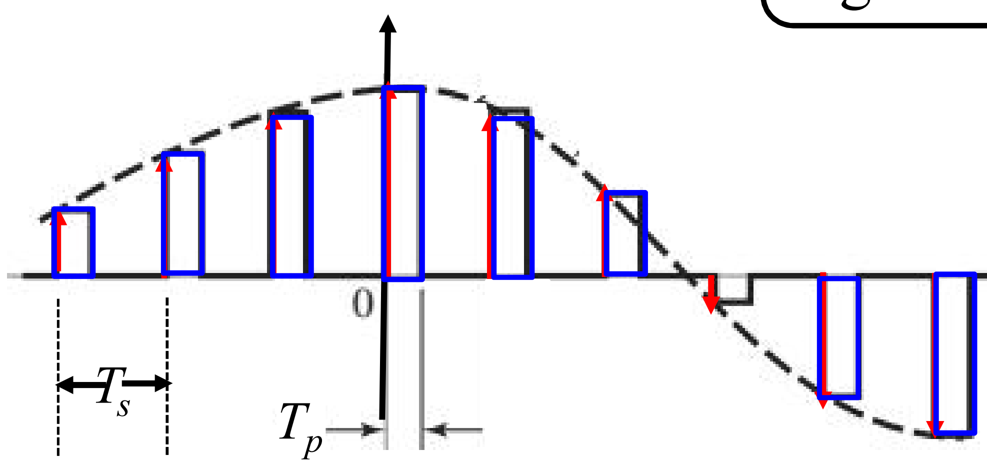


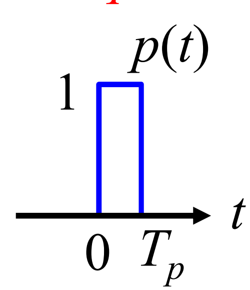
Practical Signal Reconstruction (Interpolation)



Why are these two different?

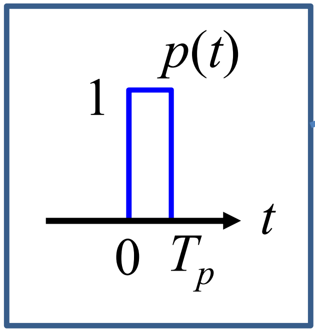
Why do we need equalizer?





Practical Signal Reconstruction (Interpolation)

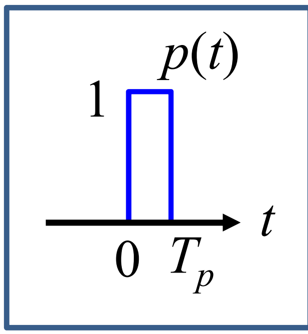
Let's find the FT response of the rectangular reconstruction pulse



$$p(t) = \Pi\left(\frac{t - 0.5T_p}{T_p}\right)$$

Practical Signal Reconstruction (Interpolation)

Let's find the FT response of the rectangular reconstruction pulse



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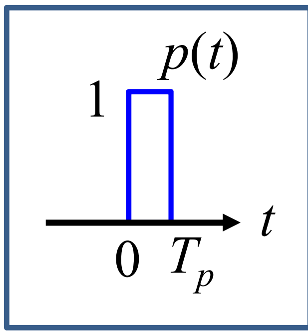

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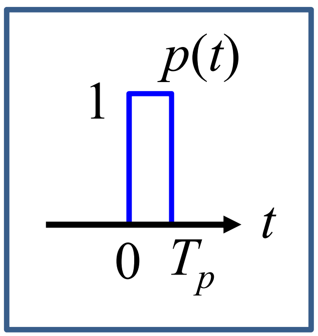
$$\Pi\left(\frac{t}{\tau}\right) \Leftrightarrow \tau \operatorname{sinc}(\pi f \tau)$$

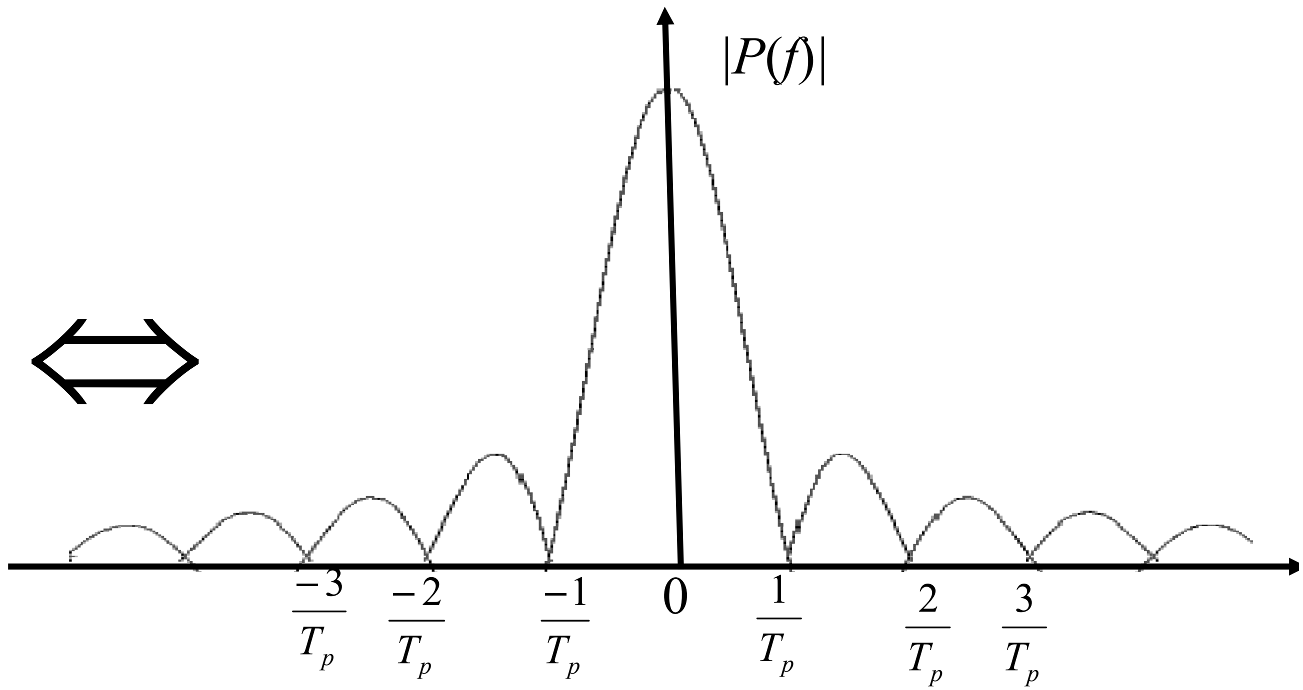
$$g(t - t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$$

Practical Signal Reconstruction (Interpolation)

rectangular
reconstruction pulse



$\Leftrightarrow$



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rectangular
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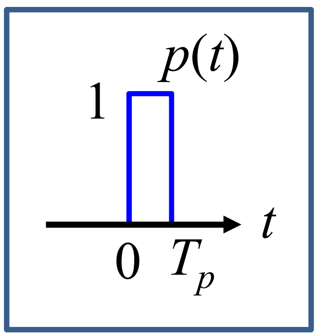

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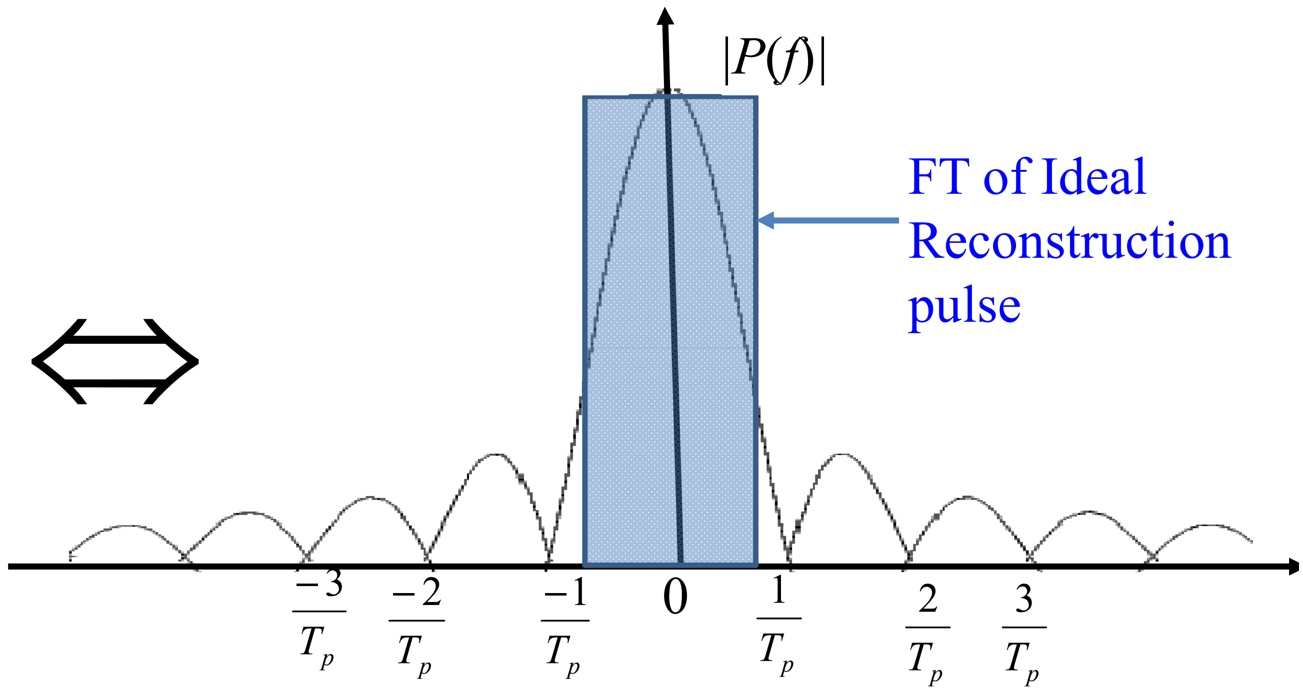

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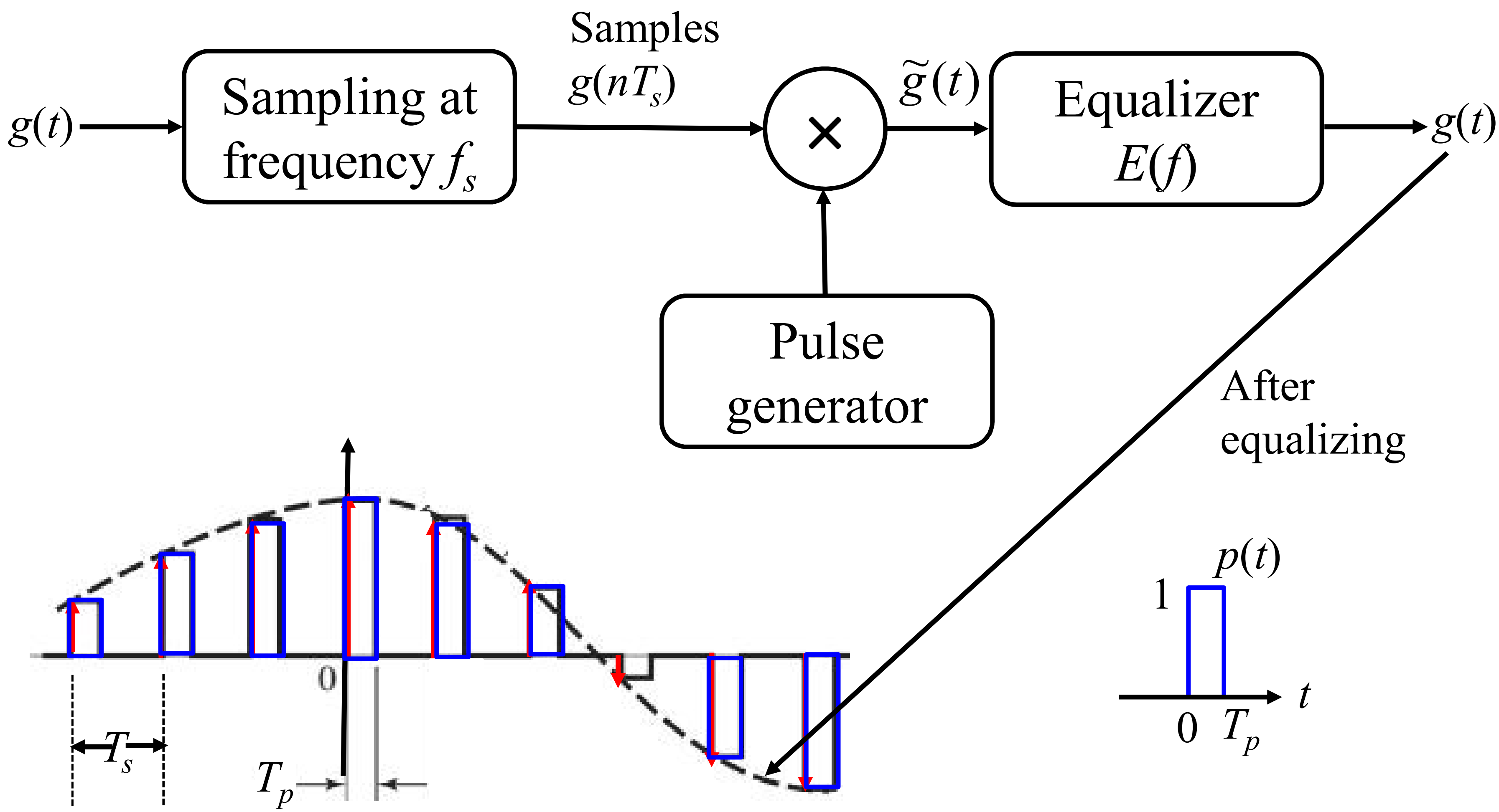
Practical Signal Reconstruction (Interpolation)



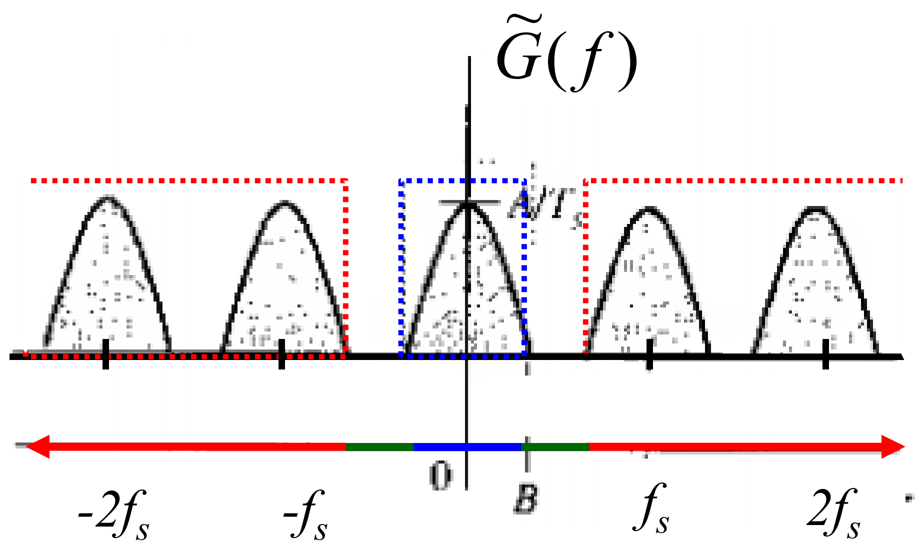
Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

$$G(f) = E(f)\tilde{G}(f)$$
$$= E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$$

Equalizer

$$E(f) = \begin{cases} T_s / P(f) & |f| \le B \\ \text{Flexible} & B < |f| \le f_s - B \\ 0 & |f| \ge f_s - B \end{cases}$$



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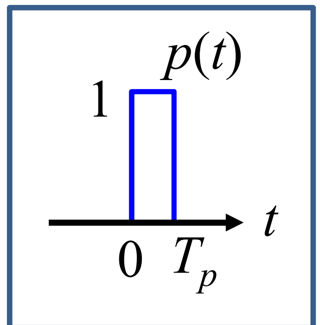

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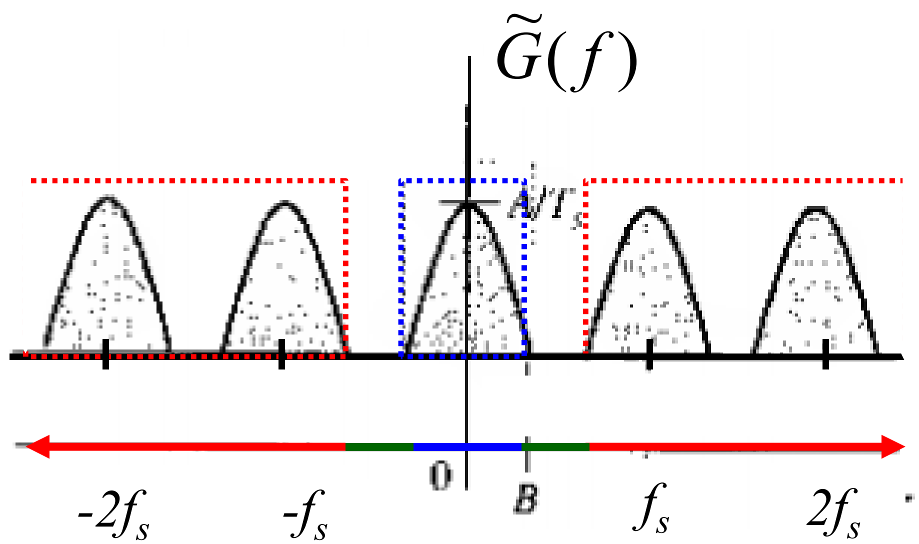

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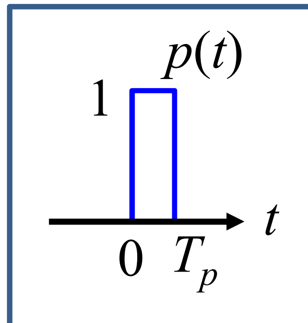

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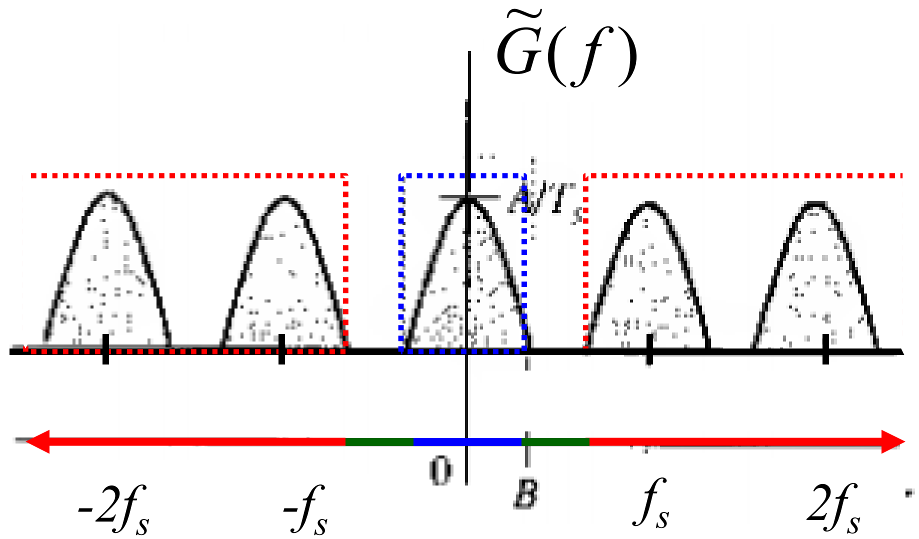

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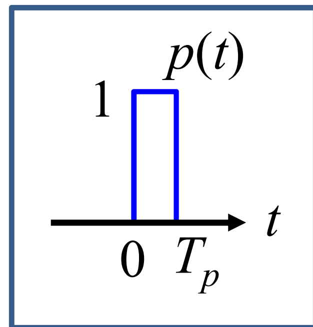

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